

教材編號：U220

程式設計概論-補充教材



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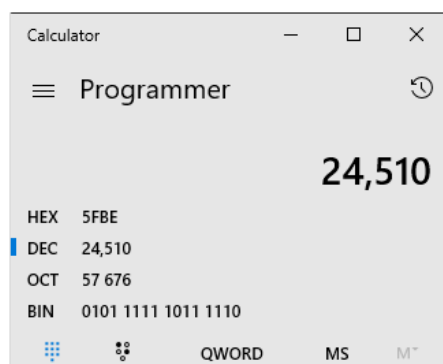
# Part-1 電腦基本概念

## 問題 1.

數字 **24510** 的二進位碼(Binary Code)、十六進位碼(Heximal Code)

解答：

打開 **Calculator**，計算機類型切換為 **Programmer** 輸入 **24150**



HEX(十六進位碼)： **5FB E**

BIN(二進位碼)： **0101 1111 1011 1110**

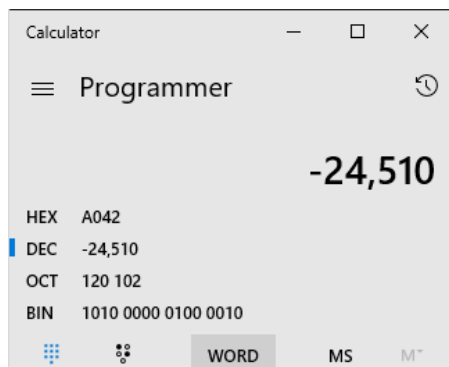
## 問題 2.

數字 **-24510** 的二進位碼(Binary Code)、十六進位碼(Heximal Code)

解答：

打開 **Calculator**，計算機類型切換為 **Programmer** 輸入 **-24510**

將位元長度 **QWORD(8 bytes)**改為 **WORD(2 bytes)**



HEX(十六進位碼)： **A042**

BIN(二進位碼)： **1010 0000 0100 0010**

**問題 3.**

請說明電腦中 CPU、RAM 的原理與功能。

**解答：**

**CPU(Central Process Unit)中央處理單元**，內部分為兩個主要部分

**1. ALU(Arithmetic Logic Unit)運算邏輯單元**

兩個主要的運算

**AU(Arithmetic Unit)算術單元**，負責加、減、乘、除運算

**LU(Logic Unit)邏輯單元**，負責邏輯判斷，比較運算(比較兩個數值的大小)，真(True)、偽(False)、NOT、AND、OR 等邏輯運算。

**2. CU(Control Unit)控制單元**

負責控制執行傳遞指令通知 ALU 執行命令，以及 ALU 與記憶體之間的溝通，通知暫存器(Register)將資料寫入記憶體、或是由記憶體終將資料搬到 ALU 運算。

**RAM(Random Access Memory)隨機存取記憶體**

電腦中儲存程式執行中需要用的指令或資料。

**問題 4.**

在一個 **100Mbps** 的網路速度環境中，要傳一個 **1.2GB** 檔案大小的 ISO 檔，需要多少時間(假設網路速度都是穩定的，檔案也沒有任何問題，忽略傳輸以外封包加解密等問題)。

**解答：**

100Mbps 是指每秒鐘傳遞(大約) 100 Mega bit，一秒鐘傳輸約  $100 / 8 = 12.5$  MB， $1.2\text{GB} =$

$1.2 \times 1024 \text{ MB} = 1228.8\text{MB}$

$1228.8 / 12.5 = 98.304$

大約 98 秒

(實際上  $100\text{Mbps} = 100 \times 10^6 \text{ bit/sec}$ )

## 問題 5.

下表為 ASCII 表

| 二進位       | 十進位 | 十六進位 | 圖形      | 二進位       | 十進位 | 十六進位 | 圖形 | 二進位       | 十進位 | 十六進位 | 圖形 |
|-----------|-----|------|---------|-----------|-----|------|----|-----------|-----|------|----|
| 0010 0000 | 32  | 20   | (space) | 0100 0000 | 64  | 40   | @  | 0110 0000 | 96  | 60   | `  |
| 0010 0001 | 33  | 21   | !       | 0100 0001 | 65  | 41   | A  | 0110 0001 | 97  | 61   | a  |
| 0010 0010 | 34  | 22   | "       | 0100 0010 | 66  | 42   | B  | 0110 0010 | 98  | 62   | b  |
| 0010 0011 | 35  | 23   | #       | 0100 0011 | 67  | 43   | C  | 0110 0011 | 99  | 63   | c  |
| 0010 0100 | 36  | 24   | \$      | 0100 0100 | 68  | 44   | D  | 0110 0100 | 100 | 64   | d  |
| 0010 0101 | 37  | 25   | %       | 0100 0101 | 69  | 45   | E  | 0110 0101 | 101 | 65   | e  |
| 0010 0110 | 38  | 26   | &       | 0100 0110 | 70  | 46   | F  | 0110 0110 | 102 | 66   | f  |
| 0010 0111 | 39  | 27   | '       | 0100 0111 | 71  | 47   | G  | 0110 0111 | 103 | 67   | g  |
| 0010 1000 | 40  | 28   | (       | 0100 1000 | 72  | 48   | H  | 0110 1000 | 104 | 68   | h  |
| 0010 1001 | 41  | 29   | )       | 0100 1001 | 73  | 49   | I  | 0110 1001 | 105 | 69   | i  |
| 0010 1010 | 42  | 2A   | *       | 0100 1010 | 74  | 4A   | J  | 0110 1010 | 106 | 6A   | j  |
| 0010 1011 | 43  | 2B   | +       | 0100 1011 | 75  | 4B   | K  | 0110 1011 | 107 | 6B   | k  |
| 0010 1100 | 44  | 2C   | ,       | 0100 1100 | 76  | 4C   | L  | 0110 1100 | 108 | 6C   | l  |
| 0010 1101 | 45  | 2D   | -       | 0100 1101 | 77  | 4D   | M  | 0110 1101 | 109 | 6D   | m  |
| 0010 1110 | 46  | 2E   | .       | 0100 1110 | 78  | 4E   | N  | 0110 1110 | 110 | 6E   | n  |
| 0010 1111 | 47  | 2F   | /       | 0100 1111 | 79  | 4F   | O  | 0110 1111 | 111 | 6F   | o  |
| 0011 0000 | 48  | 30   | 0       | 0101 0000 | 80  | 50   | P  | 0111 0000 | 112 | 70   | p  |
| 0011 0001 | 49  | 31   | 1       | 0101 0001 | 81  | 51   | Q  | 0111 0001 | 113 | 71   | q  |
| 0011 0010 | 50  | 32   | 2       | 0101 0010 | 82  | 52   | R  | 0111 0010 | 114 | 72   | r  |
| 0011 0011 | 51  | 33   | 3       | 0101 0011 | 83  | 53   | S  | 0111 0011 | 115 | 73   | s  |
| 0011 0100 | 52  | 34   | 4       | 0101 0100 | 84  | 54   | T  | 0111 0100 | 116 | 74   | t  |
| 0011 0101 | 53  | 35   | 5       | 0101 0101 | 85  | 55   | U  | 0111 0101 | 117 | 75   | u  |
| 0011 0110 | 54  | 36   | 6       | 0101 0110 | 86  | 56   | V  | 0111 0110 | 118 | 76   | v  |
| 0011 0111 | 55  | 37   | 7       | 0101 0111 | 87  | 57   | W  | 0111 0111 | 119 | 77   | w  |
| 0011 1000 | 56  | 38   | 8       | 0101 1000 | 88  | 58   | X  | 0111 1000 | 120 | 78   | x  |
| 0011 1001 | 57  | 39   | 9       | 0101 1001 | 89  | 59   | Y  | 0111 1001 | 121 | 79   | y  |
| 0011 1010 | 58  | 3A   | :       | 0101 1010 | 90  | 5A   | Z  | 0111 1010 | 122 | 7A   | z  |
| 0011 1011 | 59  | 3B   | ;       | 0101 1011 | 91  | 5B   | [  | 0111 1011 | 123 | 7B   | {  |
| 0011 1100 | 60  | 3C   | <       | 0101 1100 | 92  | 5C   | \  | 0111 1100 | 124 | 7C   |    |
| 0011 1101 | 61  | 3D   | =       | 0101 1101 | 93  | 5D   | ]  | 0111 1101 | 125 | 7D   | }  |
| 0011 1110 | 62  | 3E   | >       | 0101 1110 | 94  | 5E   | ^  | 0111 1110 | 126 | 7E   | ~  |
| 0011 1111 | 63  | 3F   | ?       | 0101 1111 | 95  | 5F   | _  |           |     |      |    |

請查出 Hello world! 的二進位碼、十進位碼、十六進位碼

## 解答：

十六進位碼： 48 65 6C 6C 6F 20 77 6F 72 6C 44 21

十進位碼： 72 101 108 108 111 32 119 111 114 108 100 33

二進位碼： 0100 1000 0110 0101 0110 1100 0100 1100 0110 1111 0010 0000  
0111 0111 0110 1111 1001 0010 0110 1100 0100 0100 0010 0001

## 問題 6.

請將以下二進位碼轉以 ASCII 轉為文字

0101 0111 0110 1000 0110 0001 0111 0100 0010 0111 0111 0011 0010 0000  
0111 1001 0110 1111 0111 0101 0111 0010 0010 0000 0110 1110 0110 0001  
0110 1101 0110 0101 0011 1111

解答：

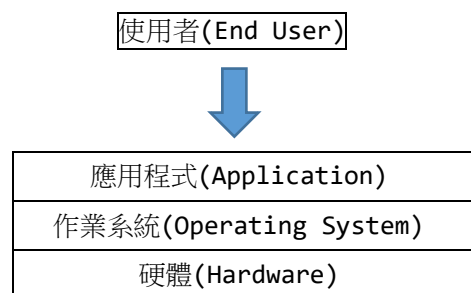
0101 0111(二進位碼) => 57(十六進位碼) => w(圖形)  
0110 1000(二進位碼) => 68(十六進位碼) => h(圖形)  
0110 0001(二進位碼) => 61(十六進位碼) => a(圖形)  
0111 0100(二進位碼) => 74(十六進位碼) => t(圖形)  
0010 0111(二進位碼) => 27(十六進位碼) => ' (圖形)  
0111 0011(二進位碼) => 73(十六進位碼) => s(圖形)  
0010 0000(二進位碼) => 20(十六進位碼) => (圖形)  
0111 1001(二進位碼) => 79(十六進位碼) => y(圖形)  
0110 1111(二進位碼) => 6F(十六進位碼) => o(圖形)  
0111 0101(二進位碼) => 75(十六進位碼) => u(圖形)  
0111 0010(二進位碼) => 72(十六進位碼) => r(圖形)  
0010 0000(二進位碼) => 20(十六進位碼) => (圖形)  
0110 1110(二進位碼) => 6E(十六進位碼) => n(圖形)  
0110 0001(二進位碼) => 61(十六進位碼) => a(圖形)  
0110 1101(二進位碼) => 6D(十六進位碼) => m(圖形)  
0110 0101(二進位碼) => 65(十六進位碼) => e(圖形)  
0011 1111(二進位碼) => 3F(十六進位碼) => ?(圖形)

#### 問題 7.

請說明作業系統的運作原理

解答：

作業系統(OS, Operating System)，負責分配系統資源，提供應用程式所需的環境。



**問題 8.**

查詢自己電腦的 IP 位址

**解答：**

點選工具列中的「**快速搜尋**」，鍵入「**cmd**」，選取「**命令提示字元(Command Prompt)**」，輸入「**IPCONFIG**」按 **Enter** 鍵

**問題 9.**

建立一個資料夾、分享給區域網路中的其他使用者

**解答：**

1. 開啟「**檔案總管**」，選取「**Windows(C:)**」，在畫面中按**右鍵**，選取「**新增(New)**」，選取「**資料夾(Folder)**」名稱上輸入「**Data**」
2. 在「**Data**」資料夾上按**右鍵**，選取「**屬性(Properties)**」，切換到「**共用(Share)**」頁籤，點擊「**進階共用(Advanced)**」按鈕。
3. 核取「**共用此資料夾**」，按**確定**完成設定，按**關閉**關閉視窗

## Part-2 C#程式語言

### 問題 1.

使用 Visual Studio 建立 Hello world 程式

解答：

```
Console.WriteLine("Hello world!");
```

### 問題 2.

使用畫面上輸入姓名(name)，顯示”Hello, (輸入姓名)!”。

解答：

```
string s;  
Console.Write("Your name:");  
s = Console.ReadLine();  
Console.WriteLine($"Hello, {s}");
```

### 問題 3.

利用 sizeof(), MinValue, MaxValue 找出 byte, short, int, long, float, double, decimal 型別佔記憶體空間的 Byte 數，以及最小值、最大值。

解答：

```
Console.WriteLine("byte:size={0}, {1}~{2}",  
    sizeof(byte), byte.MinValue, byte.MaxValue);  
Console.WriteLine("short:size={0}, {1}~{2}",  
    sizeof(short), short.MinValue, short.MaxValue);  
Console.WriteLine("int:size={0}, {1}~{2}",  
    sizeof(int), int.MinValue, int.MaxValue);  
Console.WriteLine("long:size={0}, {1}~{2}",
```



```
sizeof(long), long.MinValue, long.MaxValue);
```

**問題 4.**

利用 C#，顯示-24150，24150 這兩個數字的十六進位碼。

解答：

```
Console.WriteLine("{0}=>{0:X8}, {1}=>{1:X8}", -24150, 24150);
```

**問題 5.**

宣告變數 a，b 分別為 short a; int b; 指定 b = 100000 再將 b 的值指定給 a，顯示 a，b 的結果(十進位制)以及 16 進位碼。

解答：

```
short a;  
int b;  
b = 100000;  
a = (short)b;  
Console.WriteLine($"a={a}:{a:X8},b={b}:{b:X8}");
```

**問題 4.**

宣告變數 n 為整數，int n; 以及 s 為字串，string s; 設定 n = 1234; 並將 n 的結果給 s。

解答：

```
int n;  
string s;  
n = 1234;  
s = n.ToString();  
Console.WriteLine($"n={n}, s={s}");
```

**問題 5.**

詢問使用者姓名、生日，

顯示“(姓名)，您好，您出生於 (年/月/日)，今年(年齡)歲 。”

解答：

```
string s, name;
DateTime birthdate;
TimeSpan diff;
double age;

//輸入姓名
Console.Write("姓名:");
name = Console.ReadLine();

//輸入年齡
Console.Write("輸入出生年月日:");
s = Console.ReadLine();
birthdate = DateTime.Parse(s);
diff = DateTime.Today.Subtract(birthdate);
age = diff.Days / 365.25;
Console.WriteLine($"{name},生日:{birthdate:yyyy/MM/dd},{age:0.00}歲");
```

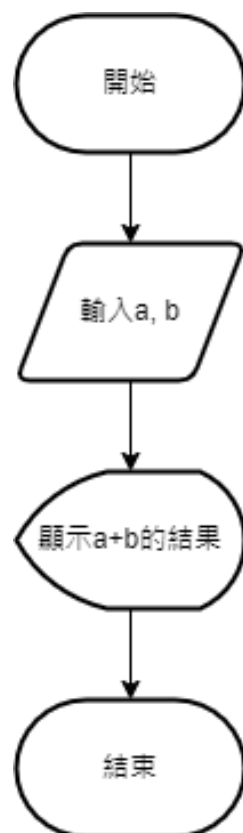
**問題 6.**

輸入  $a$ ,  $b$  兩個數字，將兩數相加結果顯示在畫面上，完成流程圖(Flowchart)、偽碼(Pseudocode)以及 C# 程式碼，並執行測試結果。

解答：

**Pseudocode**

輸入  $a$   
輸入  $b$   
計算  $a + b$  的結果

**流程圖****程式碼**

```
string s;  
int a, b;  
  
//輸入 a  
Console.Write("a=");  
s = Console.ReadLine();  
a = int.Parse(s);  
  
//輸入 b  
Console.Write("b=");  
s = Console.ReadLine();  
b = int.Parse(s);  
  
//計算 a +b 的結果  
Console.WriteLine($"{a} + {b} = {a + b}");
```

**問題 7.**

輸入成績，判斷是否及格(低於 60 不及格，高於或等於 60 及格)，完成流程圖(Flowchart)、偽碼(Pseudocode)以及 C#程式碼，並執行測試結果。

解答：

**Pseudocode**

輸入成績(score)

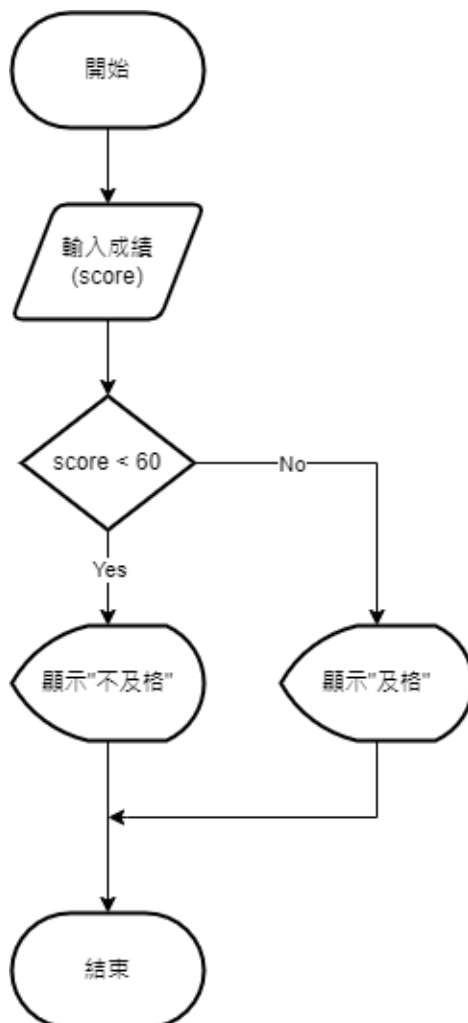
IF score < 60 THEN

    顯示 “不及格”

ELSE

    顯示 “及格”

END IF

**流程圖****程式碼**

```
int score;
string s;

//輸入成績(score)
Console.Write("成績:");
s = Console.ReadLine();
score = int.Parse(s);

//判斷是否及格
if(score < 60)
    Console.WriteLine("不及格");
else
    Console.WriteLine("及格");
```

**問題 8.**

輸入成績，判斷等級，低於 60(不含等於 60)，顯示 “C-Class”，

60 以上(含等於 60)，80 以下(不含等於 80)，顯示 “B-Class”，

80 以上(包含等於 80)，顯示 A-Class。

完成流程圖(Flowchart)、偽碼(Pseudocode)以及 C#程式碼，並執行測試結果。

解答：

**Pseudocode**

輸入成績(score)

IF score < 60 THEN

    顯示 “C-Class”

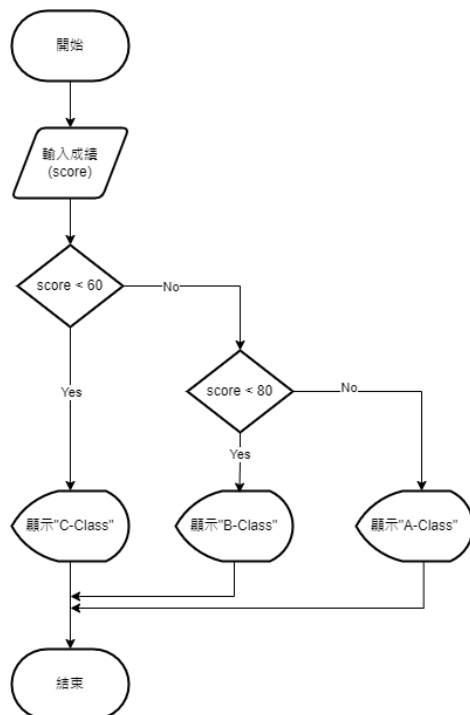
ELSEIF score < 80 THEN

    顯示 “B-Class”

ELSE

    顯示 “A-Class”

END IF

**流程圖****程式碼**

```
int score;
string s;

//輸入成績(score)
Console.Write("成績:");
s = Console.ReadLine();
score = int.Parse(s);

if (score < 60)
    Console.WriteLine("C-Class");
else if(score < 80)
    Console.WriteLine("B-Class");
else
    Console.WriteLine("A-Class");
```

## 問題 9.

所得稅級距公式如下

107 年度綜合所得稅速算公式一覽表（單位：新台幣元）

| 綜合所得淨額    |   |           |   | 稅率  |   | 累進差額    |   | 應納稅額 |
|-----------|---|-----------|---|-----|---|---------|---|------|
| 0         | ~ | 540,000   | × | 5%  | - | 0       | = |      |
| 540,001   | ~ | 1,210,000 | × | 12% | - | 37,800  | = |      |
| 1,210,001 | ~ | 2,420,000 | × | 20% | - | 134,600 | = |      |
| 2,420,001 | ~ | 4,530,000 | × | 30% | - | 376,600 | = |      |
| 4,530,001 | ~ | 以上        | × | 40% | - | 829,600 | = |      |

輸入淨所得額，計算應納稅額，完成流程圖(Flowchart)、偽碼(Pseudocode)以及 C#程式碼，並執行測試結果。

解答：

## Pseudocode

輸入年所得收入(income)

IF income <= 540000 THEN

    所得稅(tax) = income \* 0.05

ELSEIF income <= 1210000 THEN

    tax = income \* 0.12 - 37800

ELSEIF income <= 2420000 THEN

    tax = income \* 0.2 - 134600

ELSEIF income <= 4530000 THEN

    tax = income \* 0.3 - 376600

ELSE

    tax = income \* 0.4 - 829600

END IF

顯示 年所得收入(income)，應繳所得稅(tax)

## 程式碼

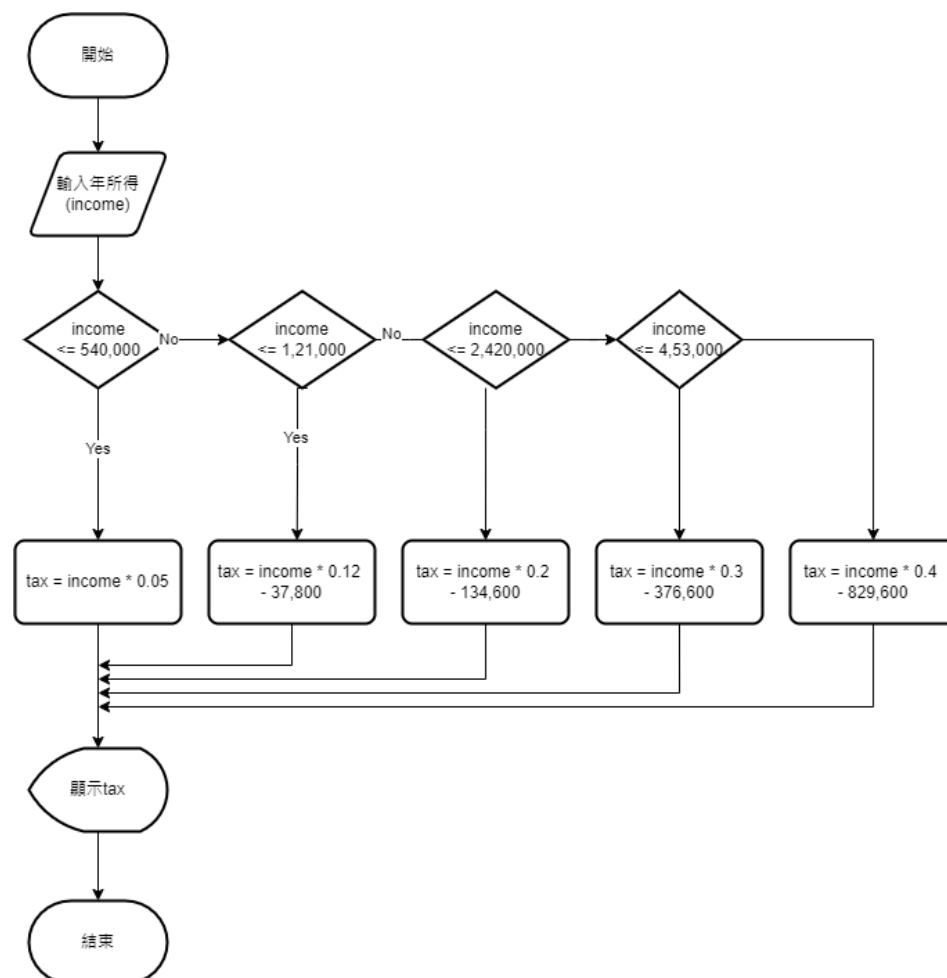
```
string s;
int income, tax;

//輸入年所得收入(income)
Console.Write("年所得收入:");
s = Console.ReadLine();
income = int.Parse(s);
```

```
if (income <= 540000)
    tax = (int)(income * 0.05 + 0.5);
else if (income <= 1210000)
    tax = (int)(income * 0.12 - 37800 + 0.5);
else if (income <= 2420000)
    tax = (int)(income * 0.2 - 134600 + 0.5);
else if (income <= 4530000)
    tax = (int)(income * 0.3 - 376600 + 0.5);
else
    tax = (int)(income * 0.4 - 829600 + 0.5);

//顯示 年所得收入(income), 應繳所得稅(tax)
Console.WriteLine($"年所得收入:{income:#,##0}, 應納所得稅:{tax:#,##0}");
```

流程圖



**問題 10.**

輸入月份，

**1, 3, 5, 7, 8, 10, 12** 月顯示 **大月(31 天)**；

**4, 6, 9, 11** 月顯示 **小月(30 天)**；

**2** 月顯示 **二月(28 或 29 天)**。

完成偽碼(Pseudocode)以及 C#程式碼，並執行測試結果。

**解答：**

**Pseudocode**

輸入月份(mn)

**CASE mn**

**1, 3, 5, 7, 8, 10, 12:**

顯示 “大月(31 天)”

**4, 6, 9, 11:**

顯示 “小月(30 天)”

**2:**

顯示 “二月(28 或 29 天)”

**ELSE**

顯示 “僅接受 1-12”

**END CASE**

**程式碼**

```
string s;
int mn;
//輸入月份(mn)
Console.Write("月份:");
s = Console.ReadLine();
mn = int.Parse(s);
switch(mn)
{
    case 1:
    case 3:
    case 5:
    case 7:
    case 8:
    case 10:
    case 12:
```



```
        Console.WriteLine("大月(31 天)");  
        break;  
    case 4:  
    case 6:  
    case 9:  
    case 11:  
        Console.WriteLine("小月(30 天)");  
        break;  
    case 2:  
        Console.WriteLine("二月(28 或 29 天)");  
        break;  
    default:  
        Console.WriteLine("僅接受 1-12");  
        break;  
}
```

**問題 11.**

輸入幣別(U=USD/J=JPY/E=EUR)，台幣金額，按照匯率

NT:USD = 1: 28.93，NT:JPY = 1:0.2048，NT:EUR = 1:34.05

(可依現在匯率調整數字)

計算可兌換外幣。完成流程圖(Flowchart)、偽碼(Pseudocode)以及 C#程式碼，並執行測試結果。

解答：

**Pseudocode**

輸入台幣 nt

輸入幣別 curr(U=USD/J=JPY/E=EUR)

CASE curr

“U”: amt = nt / 28.93

“J”: amt = nt / 0.2048

“E”: amt = nt / 34.05

ELSE

amt = 0

END CASE

顯示台幣金額(nt)、可兌換外幣(amt)

## 程式碼

```
string s, curr;
int nt;
decimal amt;

//輸入台幣nt
Console.Write("台幣:");
s = Console.ReadLine();
nt = int.Parse(s);

//輸入幣別curr(U = USD / J = JPY / E = EUR)
Console.Write("幣別(U=USD/J=JPY/E=EUR):");
curr = Console.ReadLine();

switch (curr)
{
    case "U":
        amt = nt / 28.93M;
        break;
    case "J":
        amt = nt / 0.2048M;
        break;
    case "E":
        amt = nt / 34.05M;
        break;
    default:
        amt = 0;
        break;
}

//顯示台幣金額(nt)、可兌換外幣(amt)
Console.WriteLine($"台幣:{nt}, 可兌換外幣:{amt}");
```

**問題 12.**

在畫面上顯示 0 到 9

解答：

程式碼

```
for(int i = 0; i<10; i++)  
    Console.WriteLine(i);
```

**問題 7.**

執行以下程式碼片段

```
for(float n=0; n<=10; n+=0.1f)  
    Console.WriteLine(n);
```

執行結果為何，和預期的是否相符，為什麼？怎麼解決這個問題？

解答：

```
0  
0.1  
0.2  
0.3  
0.4  
0.5  
0.6  
0.70000005  
0.80000001  
0.90000001  
1  
....  
9.700001  
9.800001  
9.900002  
會發生浮點運算誤差
```

**問題 13.**

畫星星，輸入數字在畫面上以 \* 印出下圖

完成偽碼(Pseudocode)以及 C#程式碼，並執行測試結果。

1.

```
*
**
***
...
**** ..... *
(n 個*)
```

2.

```
**** ..... *(n 個*)
****.....*(n-1 個*)
....
***
**
*
```

3.

```
      *
    ***
  *****
...
***** (2 * n - 1 個 *)
```

解答：

1.

**Pseudocode**

```
輸入 n
FOR i=1 TO n
  畫 i 個 "*"
  跳行
END FOR
```

## 程式碼

```
int n;  
string s;  
//輸入 n  
Console.Write("n=");  
s = Console.ReadLine();  
n = int.Parse(s);  
  
for(int i = 1; i<=n; i++)  
{  
    //畫 i 個 “*”  
    for(int j= 1; j<=i; j++)  
        Console.Write("*");  
    //跳行  
    Console.WriteLine();  
}
```

2.

## Pseudocode

輸入 n

FOR i=n TO 1

畫 i 個 “\*”

跳行

END FOR

## 程式碼

```
int n;  
string s;  
//輸入 n  
Console.Write("n=");  
s = Console.ReadLine();  
n = int.Parse(s);  
  
for (int i = n; i >= 1; i--)  
{
```

```
        //畫 i 個 "*"
        for (int j = 1; j <= i; j++)
            Console.Write("*");
            //跳行
        Console.WriteLine();
    }
```

3.

**Pseudocode**

輸入 n

FOR i=n TO 1

畫 n - i 個 空白

畫 2 \* i - 1 個 "\*"

跳行

END FOR

**程式碼**

```
int n;
string s;
//輸入 n
Console.Write("n=");
s = Console.ReadLine();
n = int.Parse(s);

for (int i = 1; i <= n; i++)
{
    //畫 n - i 個空白
    for (int j = 1; j <= n - i; j++)
        Console.Write(" ");
    //畫 2 * i - 1 個 "*"
    for (int j = 1; j <= 2 * i - 1; j++)
        Console.Write("*");
    //跳行
    Console.WriteLine();
}
```

**問題 14.**

輸入本金、利率(%)、期數，列出每一期的期數、本金、利息、本利和。  
完成流偽碼(Pseudocode)以及 C#程式碼，並執行測試結果。

解答：

**Pseudocode**

```
輸入本金(principal)
輸入利率(rate:%)
期數(n)

FOR i = 1 TO n
    利息(interest) = principal * rate%
    顯示 期數(i), 本金(principal), 利息(interest), 本金+利息
    將利息累加至本金
END FOR
```

**程式碼**

```
string s;
int principal, n;
double rate;

//輸入本金
Console.Write("本金=");
s = Console.ReadLine();
principal = int.Parse(s);

//輸入利率(%)
Console.Write("利率(%)=");
s = Console.ReadLine();
rate = double.Parse(s);

//輸入期數
Console.Write("期數=");
s = Console.ReadLine();
n = int.Parse(s);
```

```

Console.WriteLine("期數    本        金    利        息    合        計");
Console.WriteLine("==  =====  =====  =====");
for(int i = 1; i<=n; i++)
{
    int interest = (int)(principal * rate / 100 + 0.5);
    Console.WriteLine(
        "{0,4}{1,12:#,##0}{2,12:#,##0}{3,12:#,##0}",
        i, principal, interest, principal +interest);
    principal += interest;
}
Console.WriteLine("==  =====  =====  =====");

```

### 問題 15.

完成 九九乘法表

完成偽碼(Pseudocode)以及 C#程式碼，並執行測試結果。

解答：

#### Pseudocode

```

FOR i = 1 TO 9
    顯示 i * 1 = i 到 i * 9 = i
    (i=1, 顯示: 1 * 1 = 1  1 * 2 = 2   .....  1 * 9 = 9)
    (i=2, 顯示: 2 * 1 = 2  2 * 2 = 4   .....  2 * 9 = 18)
    .....
    (i=9, 顯示: 9 * 1 = 9  9 * 2 = 18   .....  9 * 9 = 81)
END FOR

```

#### 程式碼

```

for(int i =1; i<=9; i++)
{
    for (int j = 1; j <= 9; j++)
        Console.Write($"{j,1}*{i,1}={i*j,2}    ");
    Console.WriteLine();
}

```



**問題 16.**

使用者輸入數值，直到輸入 0 為止，再將所輸入的數值加總的結果顯示在畫面上。完成偽碼 (Pseudocode) 以及 C# 程式碼，並執行測試結果。

解答：

**Pseudocode****REPEAT**    輸入整數 *n*    將 *n* 累加至 *total***UNTIL** *n* = 0**程式碼**

```
string s;
int n, total = 0;
do
{
    // 輸入整數 n
    Console.Write("n(0 for end)=");
    s = Console.ReadLine();
    n = int.Parse(s);

    // 將 n 累加至 total
    total += n;
}while (n != 0);
Console.WriteLine($"小計:{total}");
```

**問題 17**

隨機產生 1 到 99 的亂數，作為答案，讓使用者猜數字，直到猜對為止

**解答****Pseudocode**

產生 1 - 99 的亂數 *ans*

**REPEAT**

    使用者猜數字 *guess*

**IF** *guess* < *ans* **THEN**

        顯示 "高一點"

**ELSE IF** *guess* > *ans* **THEN**

        顯示 "低一點"

**ELSE**

        顯示 "猜對了"

**ENDIF**

**UNTIL** 猜對了

**程式碼**

```
Random rnd = new Random();
int ans, guess;
//產生 1 - 99 的亂數 ans
ans = rnd.Next(1, 100);
do
{
    string s;
    Console.Write("猜數字(1-99):");
    s = Console.ReadLine();
    guess = int.Parse(s);

    if(guess < ans)
        Console.WriteLine("高一點");
    else if(guess > ans)
        Console.WriteLine("低一點");
    else
        Console.WriteLine("猜對了");
} while (ans!=guess);
```

**問題 18.**

隨機產生四個 1~6 的數字，利用以下的骰子計算方式顯示點數結果

1. 四個骰子的點數相同，顯示 “一色”。
2. 三個骰子的點數相同，另一個骰子點數不同，顯示 “沒點，重擲”。
3. 兩個骰子的點數相同，另外兩個骰子的點數不相同，顯示不相同的兩個骰子的點數合計。
4. 兩個骰子的點數相同，另外兩個骰子的點數也相同，取較大的兩個相同點數的合計為點數。
5. 四個骰子的點數都不同，顯示 “沒點，重擲”。

完成偽碼(Pseudocode)以及 C#程式碼，並執行測試結果。

**解答****Pseudocode**

產生四個 1 - 6 的亂數

IF 四個骰子的點數相同 THEN

顯示 “一色”

ELSEIF 三個骰子的點數相同，另一個骰子點數不同 THEN

顯示 “沒點，重擲”

ELSEIF 兩個骰子的點數相同，另外兩個骰子的點數也相同 THEN

較大的兩個相同點數的合計為點數

ELSEIF 兩個骰子的點數相同，另外兩個骰子的點數不相同 THEN

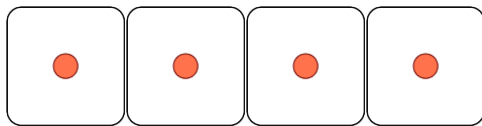
顯示不相同的兩個骰子的點數合計

ELSE

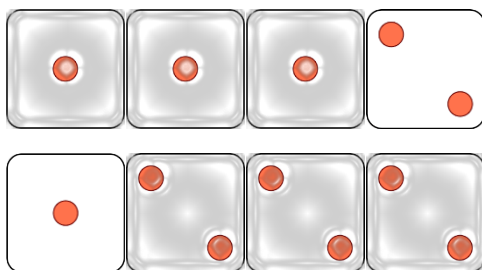
顯示 “沒點，重擲”

END IF

1. 四個骰子同樣點數(機率為:1/216)



2. 三個骰子點數相同(機率為:20/216)

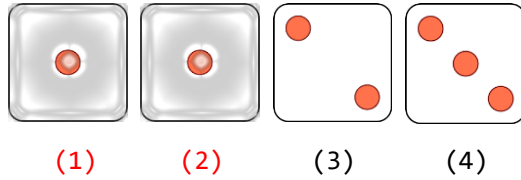


### 3. 兩個骰子點數相同

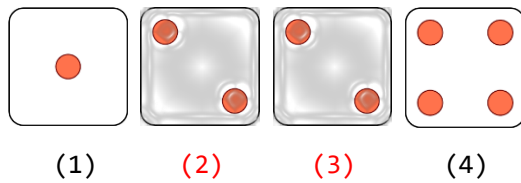
(兩個點數相同，另兩個點數也相同，機率為:15/216)

(兩個點數相同，另兩個點數不相同，機率為:60/216)

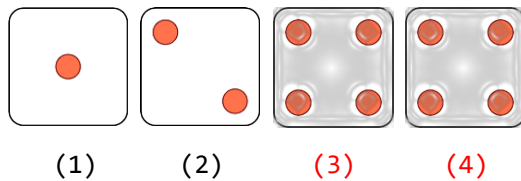
(1) = (2)



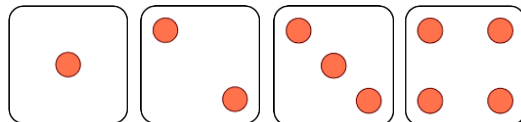
(2) = (3)



(3) = (4)



### 4. 四個骰子點數都不相同(機率為:120/216)



### 程式碼

```
Random rnd = new Random();
int[] dices = new int[4];

//產生四個1 - 6 的亂數
for (int i = 0; i < 4; i++)
    dices[i] = rnd.Next(1, 7);

//將陣列排序
Array.Sort(dices);
```

```
//由小至大顯示骰子的點數
for (int i = 0; i < dices.Length; i++)
    Console.Write("{0} ", dices[i]);
Console.WriteLine();

if (dices[0] == dices[3])
    Console.WriteLine("一色");
else if (dices[0] == dices[2] || dices[1] == dices[3])
    Console.WriteLine("沒點重擲");
else if(dices[0] == dices[1])
    Console.WriteLine("{0}點", dices[2] + dices[3]);
else if(dices[1] == dices[2])
    Console.WriteLine("{0}點", dices[0] + dices[3]);
else if (dices[2] == dices[3])
    Console.WriteLine("{0}點", dices[0] + dices[1]);
else
    Console.WriteLine("沒點重擲");
```

### 問題 19.

輸入數字 a, b 計算 a + b 的結果，避免使用者因為輸入格式錯誤而中斷程式。

### 解答

```
int a, b;
string s;
bool isincorrect;

//輸入 a
do
{
    Console.Write("a=");
    s = Console.ReadLine();
    isincorrect = int.TryParse(s, out a);
    if (!isincorrect)
        Console.WriteLine("'{0}'不是整數，請輸入整數", s);
} while (!isincorrect);
```

```
//輸入 b
do
{
    Console.Write("b=");
    s = Console.ReadLine();
    incorrect = int.TryParse(s, out b);
    if(!incorrect)
        Console.WriteLine("'{0}'不是整數，請輸入整數", s);
} while (!incorrect);

//計算 a + b 的結果
Console.WriteLine("{0} + {1} = {2}", a, b, a + b);
```

**問題 20.**

輸入十個數字，排序後，由小排到大列出，並計算出合計。

解答：

Program.cs

```
int[] ar = new int[10];
string s;

for(int i = 0; i<10; i++)
{
    Console.Write("請輸入整數#{0}:", i);
    s = Console.ReadLine();
    int.TryParse(s, out ar[i]);
}

Array.Sort(ar);

for(int i = 0; i<10; i++)
    Console.WriteLine(ar[i]);
```

**問題 21.**

高鐵票價計算，

高鐵各站票價如下

| 車站 | 南港    | 台北    | 板橋    | 桃園    | 新竹    | 苗栗    | 台中  | 彰化  | 雲林  | 嘉義    | 台南    | 左營    |
|----|-------|-------|-------|-------|-------|-------|-----|-----|-----|-------|-------|-------|
| 南港 | -     | 40    | 70    | 200   | 330   | 480   | 750 | 870 | 970 | 1,120 | 1,390 | 1,530 |
| 台北 | 40    | -     | 40    | 160   | 290   | 430   | 700 | 820 | 930 | 1,080 | 1,350 | 1,490 |
| 板橋 | 70    | 40    | -     | 130   | 260   | 400   | 670 | 790 | 900 | 1,050 | 1,320 | 1,460 |
| 桃園 | 200   | 160   | 130   | -     | 130   | 280   | 540 | 670 | 780 | 920   | 1,190 | 1,330 |
| 新竹 | 330   | 290   | 260   | 130   | -     | 140   | 410 | 540 | 640 | 790   | 1,060 | 1,200 |
| 苗栗 | 480   | 430   | 400   | 280   | 140   | -     | 270 | 390 | 500 | 640   | 920   | 1060  |
| 台中 | 750   | 700   | 670   | 540   | 410   | 270   | -   | 130 | 230 | 380   | 650   | 790   |
| 彰化 | 870   | 820   | 790   | 670   | 540   | 390   | 130 | -   | 110 | 250   | 530   | 675   |
| 雲林 | 970   | 930   | 900   | 780   | 640   | 500   | 230 | 110 | -   | 150   | 420   | 560   |
| 嘉義 | 1,120 | 1,080 | 1,050 | 920   | 790   | 640   | 380 | 250 | 150 | -     | 280   | 410   |
| 台南 | 1,390 | 1,350 | 1,320 | 1,190 | 1,060 | 920   | 650 | 530 | 420 | 280   | -     | 140   |
| 左營 | 1,530 | 1,490 | 1,460 | 1,330 | 1,200 | 1,060 | 790 | 670 | 560 | 410   | 140   | -     |

輸入起站(0-南港, 1-台北, 2-板橋....11-左營):

輸入迄站(0-南港, 1-台北, 2-板橋....11-左營):

計算票價為多少?

解答：

Program.cs

```
int[,] price =
{
    {0, 40, 70, 200, 330, 480, 750, 870, 970, 1120, 1390, 1530},
    {40, 0, 40, 160, 290, 430, 700, 820, 930, 1080, 1350, 1490},
    {70, 40, 0, 130, 260, 400, 670, 790, 900, 1050, 1320, 1460},
    {200, 160, 130, 0, 130, 280, 540, 670, 780, 920, 1190, 1330},
    {330, 290, 260, 130, 0, 140, 410, 540, 640, 790, 1060, 1200},
    {480, 430, 400, 280, 140, 0, 270, 390, 500, 640, 920, 1060},
    {750, 700, 670, 540, 410, 270, 0, 130, 230, 380, 650, 790},
    {870, 820, 790, 670, 540, 390, 130, 0, 110, 250, 530, 675},
    {970, 930, 900, 780, 640, 500, 230, 110, 0, 150, 420, 560},
    {1120, 1080, 1050, 920, 790, 640, 380, 250, 150, 0, 280, 410},
    {1390, 1350, 1320, 1190, 1060, 920, 650, 530, 420, 280, 0, 140},
    {1530, 1490, 1460, 1330, 1200, 1060, 790, 670, 560, 410, 140, 0}
};
```

```
string s;  
int n1, n2;  
  
Console.Write("起站(0-南港/1-台北/2-板橋/3-桃園/4-新竹/5-苗栗  
          /6-台中/7-彰化/8-雲林/9-嘉義/10-台南/11-左營):");  
s = Console.ReadLine();  
int.TryParse(s, out n1);  
  
Console.Write("迄站(0-南港/1-台北/2-板橋/3-桃園/4-新竹/5-苗栗  
          /6-台中/7-彰化/8-雲林/9-嘉義/10-台南/11-左營):");  
s = Console.ReadLine();  
int.TryParse(s, out n2);  
  
Console.WriteLine("票價={0}", price[n1, n2]);      }
```



## Part-3 C#物件導向

### 問題 1.

寫一個類別 Person

成員變數 private string id, name

成員方法 public void Display() 顯示 id, name

建構函式 public Person(string id, string name)設定 id, name

解答：

Person.cs

```
namespace Ap01
{
    class Person
    {
        public string id, name;
        public void Display()
        {
            Console.WriteLine("Id:{0}, Name:{1}", id, name);
        }

        public Person(string id, string name)
        {
            this.id = id;
            this.name = name;
        }
    }
}
```

Program.cs

```
using Ap01;

Person p1 = new Person("C001", "John");
p1.Display();
```

**問題 2.**

設計一類別立方體(Cube) ,

成員變數長 `private double _length, _width, _height`

成員方法體積 `public double Volume()`

成員方法表面積 `public double Area()`

建構函式 `Cube(double length, double width, double height)`

解答：

Cube.cs

```
namespace Ap02
{
    class Cube
    {
        public double _length, _width, _height;

        public Cube(double length, double width, double height)
        {
            _length = length;
            _width = width;
            _height = height;
        }

        public double Volume()
        {
            return _length * _width * _height;
        }

        public double Area()
        {
            return 2 * (_length * _width + _length * _height +
                        _width * _height);
        }
    }
}
```

Program.cs

```
using Ap02;

Cube c1 = new Cube(6, 6, 6);
Cube c2 = new Cube(12, 5, 10);
Console.WriteLine("c1: Area={0}, Volume={1}",
    c1.Area(), c1.Volume());
Console.WriteLine("c2: Area={0}, Volume={1}",
    c2.Area(), c2.Volume());
```

**問題 3.**

設計類別 BMICalculator(BMI 計算器)

成員變數 private double \_height(身高:m), \_width(體重:kg)

成員方法 public double BMI() : 體重(kg) / 身高(m)<sup>2</sup>

成員方法 public string Status():

BMI &lt; 18.5: 體重過輕

18.5 &lt;= BMI &lt; 24: 正常

BMI &gt;= 24: 體重過重

建構函式 public BMICalculator(double weight, double height)

weight: 體重(kg), height: 身高(cm)

解答：

BMICalculator.cs

```
namespace Ap03
{
    class BMICalculator
    {
        private double _height, _weight;
        public string Status()
        {
            double bmi = _weight / _height / _height;;
            if (bmivalue < 18.5)
                return string.Format($"BMI={_bmi:0.00}, 體重過輕");
            else if (bmivalue < 24)
```

```
        return string.Format($"BMI={_bmi:0.00}, 體重正常");
    else
        return string.Format($"BMI={_bmi:0.00}, 體重過重");
    }

    public BMICalculator(double weight, double height)
    {
        _weight = weight;
        _height = height / 100;
    }
}
}
```

Program.cs

```
using Ap03
BMICalculator calc = new BMICalculator(100, 180);
Console.WriteLine(calc.Status());
```

#### 問題 4.

設計類別 DiceGame(骰子遊戲)

成員變數 private int[] ans = new int[4]

成員方法 public void Roll() (擲骰子)

成員方法 public int Point() (計算點數)

解答：

Dice.cs

```
namespace Ap04
{
    class DiceGame
    {
        private int[] dice = new int[4];
        public DiceGame()
```

```
{  
    Random rnd = new Random();  
}  
  
public void Roll()  
{  
    for(int i = 0; i<4; i++)  
        dice[i] = rnd.Next(1, 7);  
}  
  
public string Display()  
{  
    Array.Sort(dice);  
  
    for (int i = 0; i < dice.Length; i++)  
        Console.Write("{0} ", dice[i]);  
  
    if(dice[0] == dice[3])  
        return “一色”;  
    else if(dice[0]==dice[2] || dice[1]==dice[3])  
        return “沒點重擲”;  
    else if(dice[0] == dice[1] && dice[2] == dice[3])  
        return string.Format(“{0}點”,  
            dice[2] + dice[3]);  
    else if(dice[0] == dice[1])  
        return string.Format(“{0}點”,  
            dice[2] + dice[3]);  
    else if(dice[1] == dice[2])  
        return string.Format(“{0}點”,  
            dice[0] + dice[3]);  
    else if(dice[2] == dice[3])  
        return string.Format(“{0}點”,  
            dice[0] + dice[1]);  
    else  
        return “沒點重擲”;  
}  
}
```

Program.cs

```
using Ap04;

DiceGame g1 = new DiceGame();
for (int i = 0; i < 10; i++)
{
    g1.Roll();
    Console.WriteLine(g1.Display());
}
```

**問題 5.**

設計類別 GuessGame(猜數字遊戲)

成員變數 private int \_guess, \_ans

成員方法 public void Start() 產生 1-99 亂數(\_ans)

成員方法 public void Guess() 使用者猜 1-99 的數字(\_guess)

成員方法 public bool GotAns() 判斷使用者是否猜對

解答：

GuessGame.cs

```
namespace Ap05
{
    private int _guess, _ans;

    public void Start()
    {
        Random rnd = new Random();
        _ans = rnd.Next(1, 100);
    }

    public void Guess()
    {
        string s;
        Console.Write("猜數字(1-99):");
```

```
        s = Console.ReadLine();
        int.TryParse(s, out _guess);
    }

    public bool GotAns()
    {
        if(_guess < _ans)
            Console.WriteLine("高一點");
        else if(_guess > _ans)
            Console.WriteLine("低一點");
        else
            Console.WriteLine("猜對了");
        return _ans == _guess;
    }
}
```

Program.cs

---

```
using Ap05;
GuessGame g1 = new GuessGame();
g1.Start();
do
{
    g1.Guess();
} while (!g1.GotAns());
```

### 問題 6.

設計類別 GuessGame(猜數字遊戲-1A1B)

將問題 5 改成 1A1B 遊戲

## Part-4 視窗程式開發

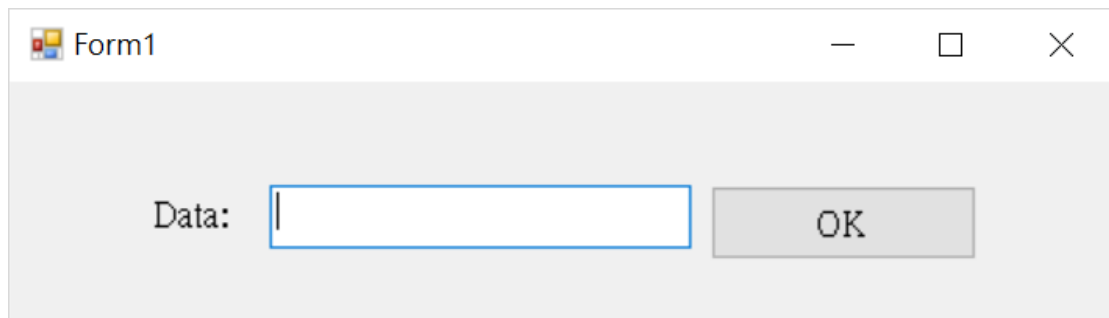
### 問題 1.

視窗中配置按鈕(Button)、標籤(Label)、文字方塊(Text Box)，按功能鍵 F5(執行)，觀察各控制項的特性。

### 問題 2.

視窗中配置按鈕(Button)、標籤(Label)、文字方塊(Text Box)，按下按鈕後，在文字方塊中出現 Hello world。

解答：



從工具箱中拖曳控制像至畫面中，並利用屬性視窗(F4)修改屬性

Label:

(Name): **lblData**

Text: **Data:**

TextBox:

(Name): **txtData**

Button:

(Name): **btnOk**

Text: **OK**

在按鈕 btnOk 上 Double Click，寫入程式碼

```
private void btnOk_Click(object sender, EventArgs e)
{
    txtData.Text = "Hello world!";
}
```



完整程式碼如下：

```
namespace Ap02
{
    public partial class Form1 : Form
    {
        public Form1()
        {
            InitializeComponent();

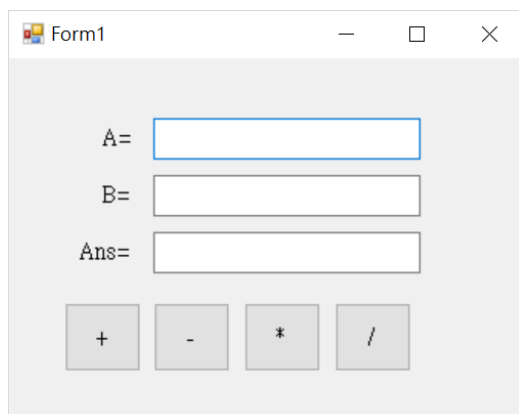
            private void btnOk_Click(object sender, EventArgs e)
            {
                txtData.Text = "Hello world!";
            }
        }
    }
}
```

### 問題 3.

文字方塊 A, B 中輸入數值，計算數值相加、減、乘、除的結果。

解答：

配置以下的控制項以及屬性



Label:

(Name): **lblA**  
Text: **A=**  
(Name): **lblB**  
Text: **B=**  
(Name): **lblAns**  
Text: **Ans=**  
TextBox:  
(Name): **txtA**  
(Name): **txtB**  
(Name): **txtAns**  
Button:  
(Name): **btnAdd**  
Text: **+**  
(Name): **btnSub**  
Text: **-**  
(Name): **btnMul**  
Text: **\***  
(Name): **btnDiv**  
Text: **/**

```
private void btnAdd_Click(object sender, EventArgs e)
{
    double a, b;
    double.TryParse(txtA.Text, out a);
    double.TryParse(txtB.Text, out b);
    txtAns.Text = (a + b).ToString();
}

private void btnSub_Click(object sender, EventArgs e)
{
    double a, b;
    double.TryParse(txtA.Text, out a);
    double.TryParse(txtB.Text, out b);
    txtAns.Text = (a - b).ToString();
}

private void btnMul_Click(object sender, EventArgs e)
```

```
{  
    double a, b;  
    double.TryParse(txtA.Text, out a);  
    double.TryParse(txtB.Text, out b);  
    txtAns.Text = (a * b).ToString();  
}  
  
private void btnDiv_Click(object sender, EventArgs e)  
{  
    double a, b;  
    double.TryParse(txtA.Text, out a);  
    double.TryParse(txtB.Text, out b);  
    txtAns.Text = (a / b).ToString();  
}
```

#### 問題 4.

輸入成績，並計算是否及格( $\geq 60$ )，並以顏色標示、及格為藍底白字、不及格為黃底紅字。

解答：

配置以下的控制項以及屬性

Label

(Name): **lblScore**

Text: 成績:

(Name): **lblResult**

Text: 結果:

TextBox

(Name): **txtScore**

(Name): **txtResult**

## Button

(Name): btnCalc

Text: 計算

```
private void btnCalc_Click(object sender, EventArgs e)
{
    int score;
    int.TryParse(txtScore.Text, out score);
    if(score < 60)
    {
        txtResult.Text = "不及格";
        txtResult.BackColor = Color.LightYellow;
        txtResult.ForeColor = Color.Red;
    }
    else
    {
        txtResult.Text = "及格";
        txtResult.BackColor = Color.Blue;
        txtResult.ForeColor = Color.White;
    }
}
```

## 問題 5.

選擇幣別(美金、日圓、歐元)，台幣金額，按照匯率

NT:USD = 1: 30.05，NT:JPY = 1:0.2785，NT:EUR = 1:32.55

(可依現在匯率調整數字)

計算可兌換外幣。利用視窗程式增進介面的親和性。

## 解答

## Label

(Name): **lblCurrency**

Text: 幣別

(Name): **lblNT**

Text: 台幣

(Name): **lblExchange**

Text: 可兌換金額

## ComboBox

(Name): **cmbCurr**

Items: 美金

日圓

歐元

## TextBox

(Name): **txtNT**

(Name): **txtAmount**

## Button

(Name): **btnExchange**

Text: 可兌換金額

```
private void btnExchange_Click(object sender, EventArgs e)
{
    int nt;
    decimal amount;
    int.TryParse(txtNT.Text, out nt);

    switch(cmbCurr.Text)
    {
        case "美金":
            amount = nt / 30.05M;
            break;
        case "日圓":
            amount = nt / 0.2756M;
            break;
        case "歐元":
            amount = nt / 32.15M;
```

```
        break;
    default:
        amount = 0;
        break;
    }
    txtAmount.Text = amount.ToString("#,##0.00");
}
```

### 問題 6.



讓使用者點選欲購買項目，按計算後加總合計金額。

### 解答

GroupBox:

(name): **grpMeal**

Text: **餐點**

RadioButton

(Name): **rdoNo1**

Text: **No1(95)**

(Name): **rdoNo2**

Text: **No2(105)**

(Name): **rdoNo3**

Text: **No3(120)**

CheckBox

(Name): **chkCream**

Text: **Ice Cream(15)**

(Name): **chkPie**

Text: **Apple Pie(25)**

Label

(Name): **lblTotal**

Text: 小計:

Button

(Name): **btnCalc**

Text: 計算

```
private void btnCalc_Click(object sender, EventArgs e)
{
    int total = 0;

    if (rdoNo1.Checked)
        total += 95;
    else if (rdoNo2.Checked)
        total += 105;
    else if (rdoNo3.Checked)
        total += 120;

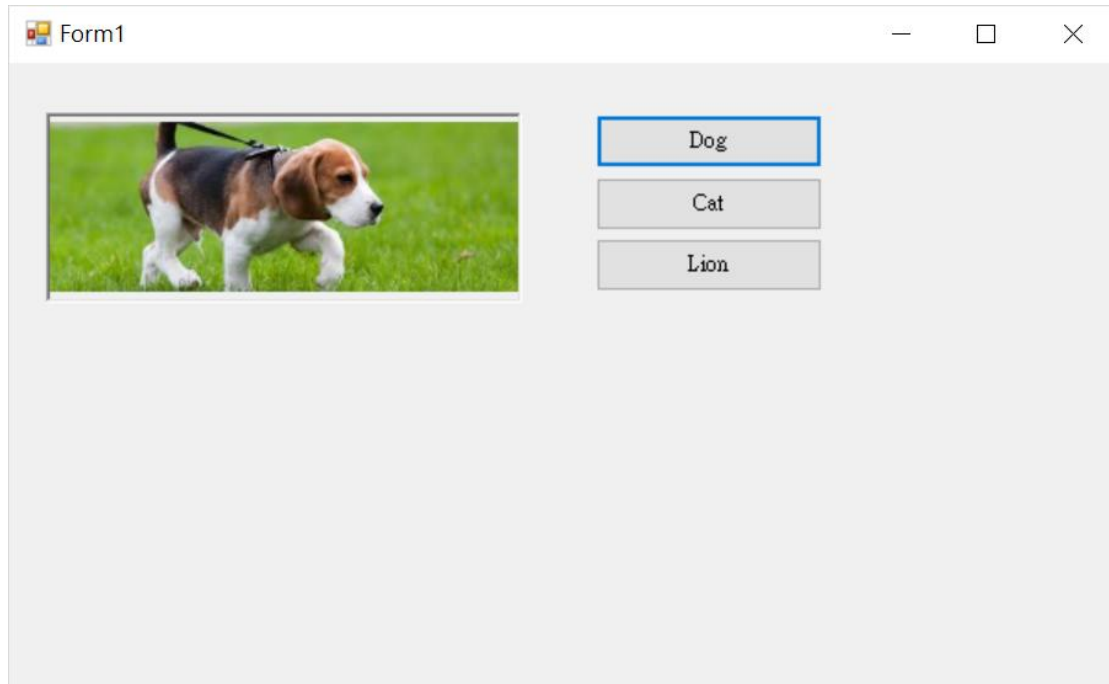
    if (chkCream.Checked)
        total += 15;

    if (chkPie.Checked)
        total += 25;

    txtTotal.Text = total.ToString();
}
```

### 問題 7.

利用檔案相對路徑載入圖片。



解答：

PictureBox

(Name): **pct1**

Button

(Name): **btnDog**

Text: **Dog**

(Name): **btnCat**

Text: **Cat**

(Name): **btnLion**

Text: **Lion**

```
private void btnDog_Click(object sender, EventArgs e)
{
    pct1.Image = Image.FromFile("animals/dog.jpg");
}

private void btnCat_Click(object sender, EventArgs e)
{
    pct1.Image = Image.FromFile("animals/cat.jpg");
}

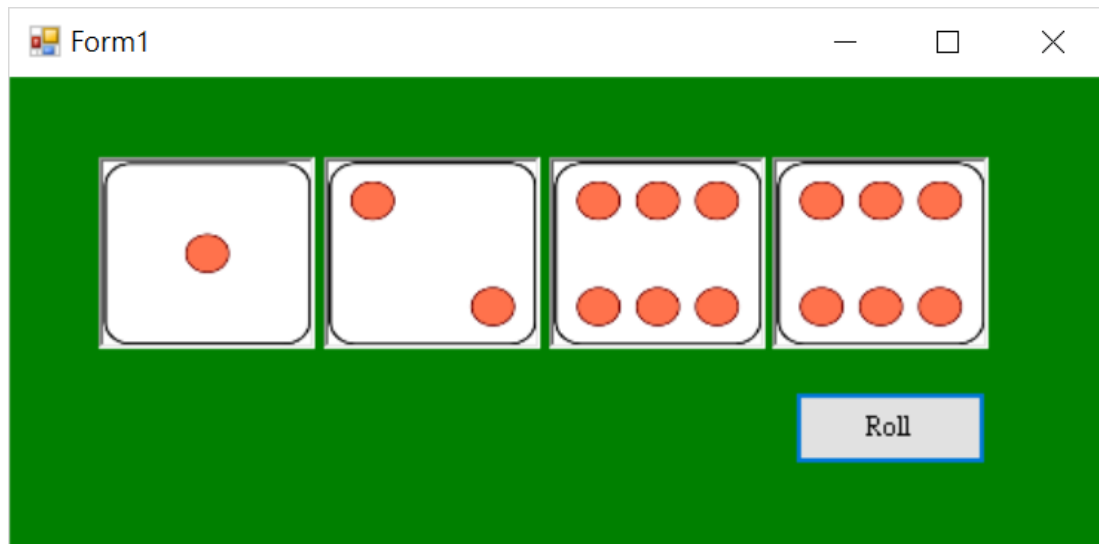
private void btnLion_Click(object sender, EventArgs e)
{
```



```
pct1.Image = Image.FromFile("animals/lion.jpg");  
}
```

**問題 8.**

擲骰子程式



按下按鈕可以隨機出現四個骰子的圖片。

**解答：**

PictureBox

```
(Name): pct1  
SizeMode: Stretch  
(Name): pct2  
SizeMode: Stretch  
(Name): pct3  
SizeMode: Stretch  
(Name): pct4  
SizeMode: Stretch
```

Button

```
(Name): btnRoll  
Text: Roll
```

```
private void btnRoll_Click(object sender, EventArgs e)  
{
```

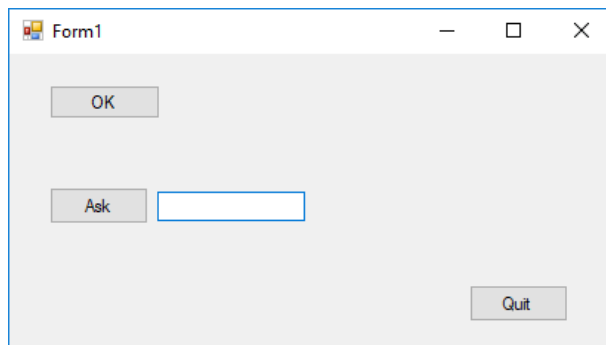
```
string[] files = {"dices/d1.png", "dices/d2.png",  
                 "dices/d3.png", "dices/d4.png",  
                 "dices/d5.png", "dices/d6.png"};  
  
Random rnd = new Random();  
int n;  
  
n = rnd.Next(0, 6);  
pct1.Image = Image.FromFile(files[n]);  
n = rnd.Next(0, 6);  
pct2.Image = Image.FromFile(files[n]);  
n = rnd.Next(0, 6);  
pct3.Image = Image.FromFile(files[n]);  
n = rnd.Next(0, 6);  
pct4.Image = Image.FromFile(files[n]);  
}
```

### 問題 9.

訊息視窗

1. 按下"OK"彈出訊息視窗顯示"Hello"。
2. 按下"Ask"顯示"Yes or No"，按下"Yes"文字方塊中寫入"Yes!"，按下"No"文字方塊中顯示"Nooooooo!"。
3. 按下"Close"詢問"確定要離開"，回答"Yes"關閉視窗，回答"No"回到程式。

解答：



Button

(Name): btnOk

Text: OK

(Name): btnAsk

Text: Ask

(Name): btnQuit

Text: Quit

TextBox

(Name): txtAns

```
private void btnOk_Click(object sender, EventArgs e)
{
    MessageBox.Show("Hello!", "Greeting",
        MessageBoxButtons.YesNo,
        MessageBoxIcon.Error);
}

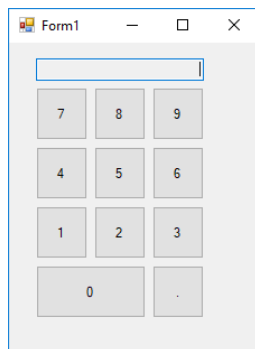
private void btnAsk_Click(object sender, EventArgs e)
{
    DialogResult ans = MessageBox.Show("Yes or No", "Ask",
        MessageBoxButtons.YesNo,
        MessageBoxIcon.Information);
    if (ans == DialogResult.Yes)
        txtAns.Text = "Yep!";
    else if (ans == DialogResult.No)
        txtAns.Text = "Noooooooooooo!";
}
```

```
private void btnQuit_Click(object sender, EventArgs e)
{
    DialogResult ans = MessageBox.Show("確定要離開", "離開",
    MessageBoxButtons.YesNo, MessageBoxIcon.Information);
    if (ans == DialogResult.Yes)
        Close();
}
```

**問題 10.****數字鍵盤**

十一個按鈕"0"、"1"、"2"、"3"、"4"、"5"、"6"、"7"、"8"、"9"、"."，按下按鈕後在文字方塊中顯示數字。

**解答：**

**Button**

(Name): btn0

Text: 0

(Name): btn1

Text: 1

(Name): btn2

Text: 2

(Name): btn3

Text: 3

(Name): btn4

Text: 4

(Name): btn5

Text: 5

(Name): btn6

Text: 6

(Name): btn7

Text: 7

(Name): btn8

Text: 8

(Name): btn9

Text: 9

(Name): btnDot

Text: .

TextBox

(Name): txtDisplay

```
private void btn_Click(object sender, EventArgs e)
{
    Button btn = (Button)sender;
    txtDisplay.Text += btn.Text;
}
```

### 問題 11.

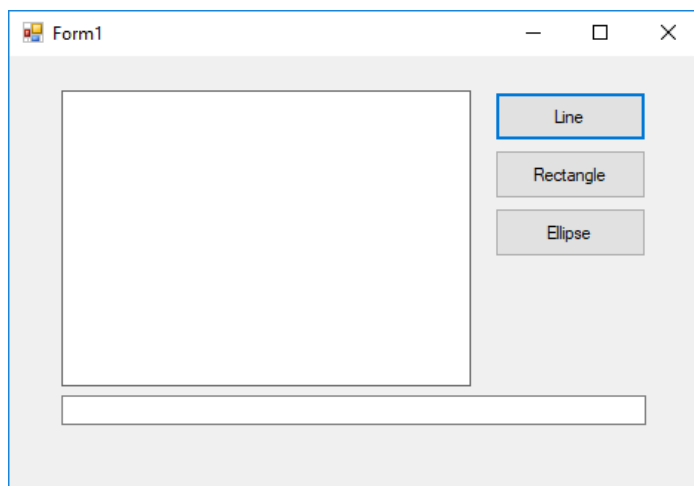
繪圖

畫線：座標(10, 10)到(110, 110)

畫方形：座標(10, 10)，寬 100，高 100

畫圓形：前述的方形中放入內接圓

解答：



PictureBox

(Name): pct1

## Button

(Name): btnLine

Text: Line

(Name): btnRectangle

Text: Rectangle

(Name): btnEllipse

Text: Ellipse

## TextBox

(Name): txtData

```
namespace Ap11
{
    public partial class Form1 : Form
    {
        int x0, y0;
        public Form1()
        {
            InitializeComponent();
        }

        private void btnLine_Click(object sender, EventArgs e)
        {
            Graphics g = pct1.CreateGraphics();
            g.DrawLine(new Pen(Color.Blue), 10, 10, 110, 110);
        }

        private void btnRectangle_Click(object sender, EventArgs e)
        {
            Graphics g = pct1.CreateGraphics();
            g.DrawRectangle(new Pen(Color.Green), 10, 10, 100, 100);
        }

        private void btnEllipse_Click(object sender, EventArgs e)
        {
            Graphics g = pct1.CreateGraphics();
            g.DrawEllipse(new Pen(Color.Red), 10, 10, 100, 100);
        }
    }
}
```

```
private void pct1_MouseDown(object sender, MouseEventArgs e)
{
    x0 = e.X;
    y0 = e.Y;
}

private void pct1_MouseMove(object sender, MouseEventArgs e)
{
    if(e.Button== MouseButtons.Left)
    {
        Graphics g = pct1.CreateGraphics();
        g.DrawLine(new Pen(Color.Black), x0, y0, e.X, e.Y);
        x0 = e.X;
        y0 = e.Y;
    }

    txtData.Text = string.Format("X={0}, Y={1}, Button={2}",
        e.X, e.Y, e.Button);
}
}
```

**問題 12.**

鍵盤操作

八個文字方塊，可以使用上下鍵控制游標可以向上或向下一個文字方塊。

解答：

```
private void txt_KeyDown(object sender, KeyEventArgs e)
{
    switch(e.KeyCode)
    {
        case Keys.Down:
        case Keys.Return:
            SendKeys.Send("{Tab}");
            break;
        case Keys.Up:
            SendKeys.Send("+{Tab}");
            break;
    }
}

private void txt_Enter(object sender, EventArgs e)
{
    TextBox txt = (TextBox)sender;
    txt.SelectAll();
}
```

**問題 13.**

計時器

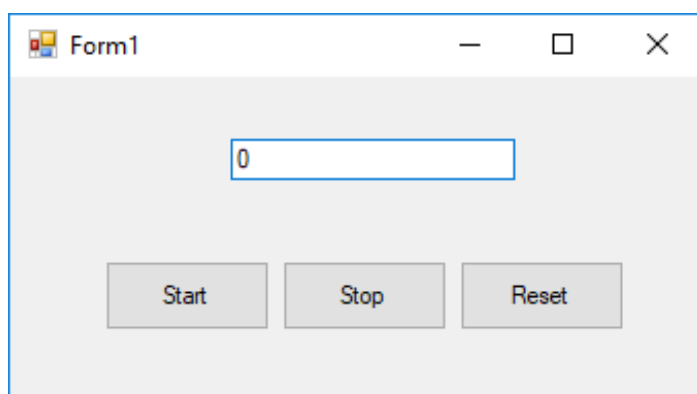
文字方塊內的起始數字為 0，

按鈕"Start"啟動計數器，每 0.1 秒加 1

按鈕"Stop"暫停計數器

按鈕"Reset"重置計數器，將文字方塊數值歸零。

解答：





## TextBox

(Name): txtN

Text: 0

## Button

(Name): btnStart

Text: Start

(Name): btnStop

Text: Stop

(Name) btnReset

Text: Reset

```
private void timer1_Tick(object sender, EventArgs e)
{
    int n;
    int.TryParse(txtN.Text, out n);
    n++;
    txtN.Text = n.ToString();
}

private void btnStart_Click(object sender, EventArgs e)
{
    timer1.Enabled = true;
}

private void btnStop_Click(object sender, EventArgs e)
{
    timer1.Enabled = false;
}

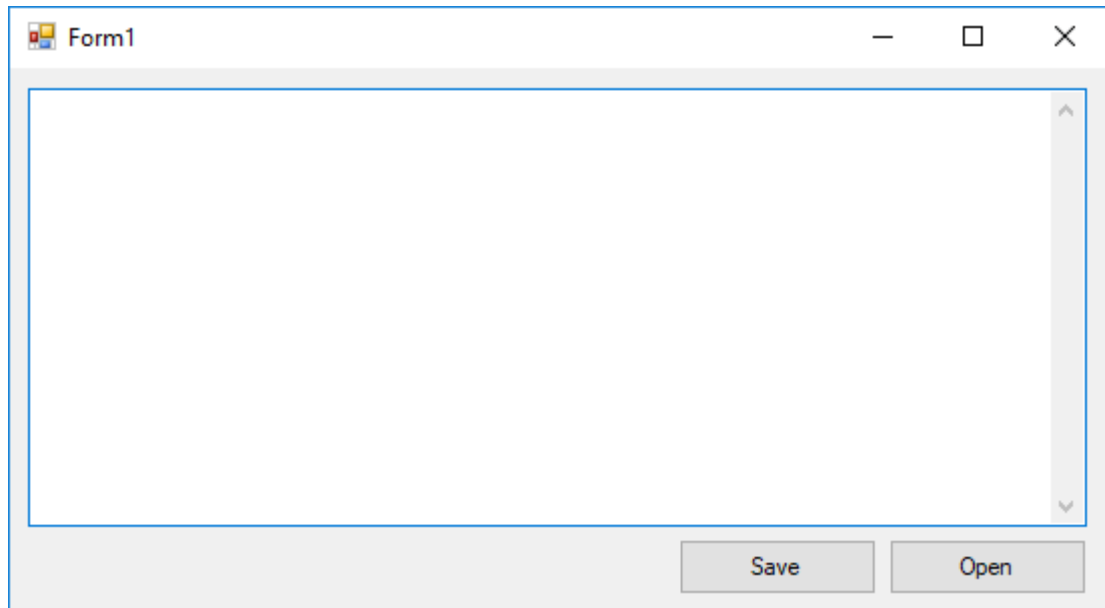
private void btnReset_Click(object sender, EventArgs e)
{
    txtN.Text = "0";
}
```

## 問題 14.

## 記事本-1

設計一介面，讓使用者輸入文字，並可存檔、讀取檔案。

解答：



TextBox

(Name):txtData

Anchor: Top, Bottom, Left, Right

Button

(Name):btnSave

Text: Save

(Name):btnOpen

Text:Open

OpenFileDialog

(Name):dlgOpen

SaveFileDialog

(Name):dlgSave

```
namespace Ap14
{
    public partial class Form1 : Form
    {
        public Form1()
        {
            InitializeComponent();
        }
    }
}
```

```
    }

    private void btnSave_Click(object sender, EventArgs e)
    {
        StreamWriter wr = new StreamWriter("C:/data/test.txt");
        wr.Write(txtData.Text);
        wr.Flush();
        wr.Close();
        MessageBox.Show("存檔成功", "存檔", MessageBoxButtons.OK,
            MessageBoxIcon.Information);
    }

    private void btnOpen_Click(object sender, EventArgs e)
    {
        StreamReader rd = new StreamReader("C:/data/test.txt");
        txtData.Clear();
        while(!rd.EndOfStream)
        {
            string s = rd.ReadLine();
            txtData.Text += s;
            txtData.Text += System.Environment.NewLine;
        }
        rd.Close();
    }
}
```

### 問題 15

設計一介面新增、修改、刪除、查詢資料表資料

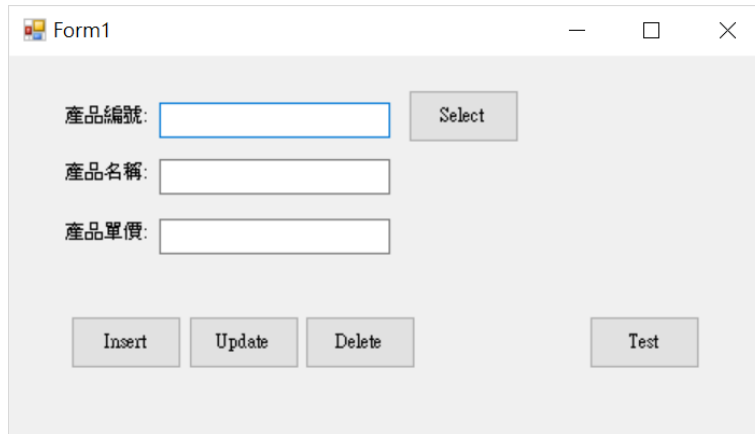
資料庫名稱：db01

資料表名稱：dbo.Product

欄位結構

| 欄位名稱      | 說明   | 資料型別         | 主索引鍵 |
|-----------|------|--------------|------|
| prdt_id   | 產品編號 | int          | Y    |
| prdt_name | 產品名稱 | nvarchar(20) |      |
| price     | 產品單價 | int          |      |

解答：



The screenshot shows a Windows Form titled "Form1". It contains three labels: "產品編號:", "產品名稱:", and "產品單價:". Each label is followed by a text box. To the right of the "產品編號:" text box is a "Select" button. Below the text boxes are four buttons: "Insert", "Update", "Delete", and "Test".

Label

(Name): lblId

Text: 產品編號

(Name): lblName

Text: 產品名稱

(Name): lblPrice

Text: 產品單價

TextBox

(Name): txtId

(Name): txtName

(Name): txtPrice

Button

(Name): btnSelect

Text: Select

(Name): btnInsert

Text: Insert

(Name): btnUpdate

Text: Update

(Name): btnDelete

Text: Delete

(Name): btnTest

Text: Test

```
namespace Ap15
{
    public partial class Form1 : Form
    {
        string connstr = @"Server=.;
                            Database=db01;
                            Trusted_Connection=true;";

        public Form1()
        {
            InitializeComponent();
        }

        private void btnTest_Click(object sender, EventArgs e)
        {
            SqlConnection cn = new SqlConnection(connstr);
            cn.Open();
            cn.Close();
            MessageBox.Show("Connect OK!", "Test",
                            MessageBoxButtons.OK, MessageBoxIcon.Information);
        }

        private void btnInsert_Click(object sender, EventArgs e)
        {
            SqlConnection cn = new SqlConnection(connstr);
            SqlCommand cmd = cn.CreateCommand();
            cmd.CommandText = @"INSERT INTO dbo.Product(prdt_id,
                                prdt_name, price)
                                VALUES(@id, @name, @price);";
            cmd.Parameters.AddWithValue("@id", txtID.Text);
            cmd.Parameters.AddWithValue("@name", txtName.Text);
            cmd.Parameters.AddWithValue("@price", txtPrice.Text);
            cn.Open();
            cmd.ExecuteNonQuery();
            cn.Close();
            MessageBox.Show("新增成功", "Insert",
                            MessageBoxButtons.OK, MessageBoxIcon.Information);
        }
    }
}
```

```
private void btnUpdate_Click(object sender, EventArgs e)
{
    SqlConnection cn = new SqlConnection(connstr);
    SqlCommand cmd = cn.CreateCommand();
    cmd.CommandText = @"UPDATE dbo.Product
        SET prdt_name = @name, price = @price
        WHERE prdt_id = @id;";
    cmd.Parameters.AddWithValue("@id", txtID.Text);
    cmd.Parameters.AddWithValue("@name", txtName.Text);
    cmd.Parameters.AddWithValue("@price", txtPrice.Text);
    cn.Open();
    cmd.ExecuteNonQuery();
    cn.Close();
    MessageBox.Show("修改成功", "Update",
        MessageBoxButtons.OK, MessageBoxIcon.Information);
}
```

```
private void btnDelete_Click(object sender, EventArgs e)
{
    SqlConnection cn = new SqlConnection(connstr);
    SqlCommand cmd = cn.CreateCommand();
    cmd.CommandText = @"DELETE FROM dbo.Product
        WHERE prdt_id = @id;";
    cmd.Parameters.AddWithValue("@id", txtID.Text);
    cn.Open();
    cmd.ExecuteNonQuery();
    cn.Close();
    txtName.Clear();
    txtPrice.Clear();
    MessageBox.Show("刪除成功", "Delete",
        MessageBoxButtons.OK, MessageBoxIcon.Information);
}
```

```
private void btnSelect_Click(object sender, EventArgs e)
{
    SqlConnection cn = new SqlConnection(connstr);
    SqlCommand cmd = cn.CreateCommand();
```

```
cmd.CommandText = @"SELECT * FROM dbo.Product
    WHERE prdt_id = @id;";
cmd.Parameters.AddWithValue("@id", txtID.Text);
cn.Open();
SqlDataReader dr = cmd.ExecuteReader();
if(dr.Read())
{
    txtName.Text = dr["prdt_name"].ToString();
    txtPrice.Text = dr["price"].ToString();
}
else
{
    txtName.Clear();
    txtPrice.Clear();
}
cn.Close();
}
}
}
```

## Part-5 關聯式資料庫

### 問題 1

資料庫名稱：db01

資料表名稱：dbo.Product

欄位結構

| 欄位名稱      | 說明   | 資料型別         | 主索引鍵 |
|-----------|------|--------------|------|
| prdt_id   | 產品編號 | int          | Y    |
| prdt_name | 產品名稱 | nvarchar(20) |      |
| price     | 產品單價 | int          |      |

以 T-SQL 語法建立

解答：

進入 SSMS => 連線到本機的 SQL Server => 按一下 New Query(新增查詢)=>鍵入：

```
CREATE DATABASE db01;
GO

USE db01;

CREATE TABLE dbo.Product
(
    prdt_id int PRIMARY KEY,
    prdt_name nvarchar(20),
    price int
);
GO
```

### 問題 2

前面的資料表中，以 T-SQL 語法新增以下三筆資料

| prdt_id | prdt_name | price |
|---------|-----------|-------|
| 1       | A-Product | 1000  |
| 2       | B-Product | 2000  |
| 3       | C-Product | 3000  |



-----

解答：

```
USE db01;
INSERT INTO dbo.Product(prdt_id, prdt_name, price)
VALUES(1, 'A-Product', 1000),
(2, 'B-Product', 2000),
(3, 'C-Product', 3000);
```

### 問題 3

以 T-SQL 將 prdt\_id = 2 的資料，修改 prdt\_name = 'New-Product', price = 9999

解答：

```
USE db01;
UPDATE dbo.Product
SET prdt_name = 'New-Product', price = 9999
WHERE prdt_id = 2;
```

### 問題 4

以 T-SQL 刪除 prdt\_id = 3 的資料

解答：

```
USE db01;
DELETE FROM dbo.Product
WHERE prdt_id = 3;
```

### 問題 5

以 T-SQL 查詢 prdt\_id = 1 的資料(所有欄位)

解答：

```
USE db01;
SELECT * FROM dbo.Product
WHERE prdt_id = 1;
```

# Part-6 Web 應用程式

## HTML

### 問題 1：

在瀏覽器中出現 Hello world!



### 解答：

```
<html>
  <head>
    <title>My first HTML</title>
  </head>
  <body style="color:blue;">
    Hello world!
  </body>
</html>
```

### 問題 2：

標題字

**Header 1**

**Header 2**

**Header 3**

**Header 4**

**Header 5**

**Header 6**

解答：

```
<html>
  <head>
    <title>HTML 2</title>
  </head>
  <body>
    <h1>Header 1</h1>
    <h2>Header 2</h2>
    <h3>Header 3</h3>
    <h4>Header 4</h4>
    <h5>Header 5</h5>
    <h6>Header 6</h6>
  </body>
</html>
```

問題 3：

跳行、空白、水平尺規、粗體、斜體、底線

Line 1  
Line 2

---

Hello, my name is **John**.  
**Underline** *Italic*

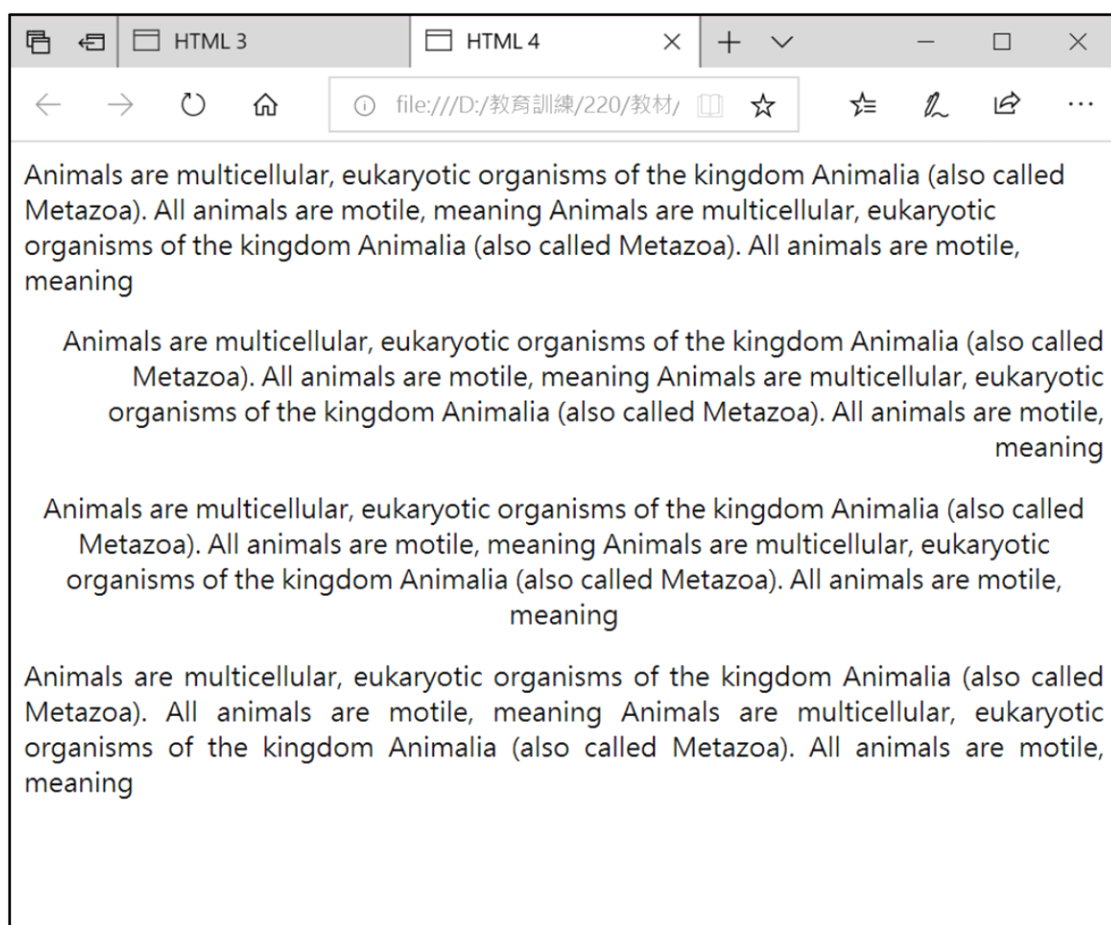
解答：

```
<html>
  <head>
    <title>HTML 3</title>
  </head>
  <body>
    Line 1<br />Line 2<br />
    <hr />
    Hello, my name is <b>John</b>.
```

```
<br />
<b>Bold</b>
<u>Underline</u>
<i>Italic</i>
<body>
</html>
```

#### 問題 4：

文字排版(靠左對齊、靠右對齊、置中對齊、分散對齊)



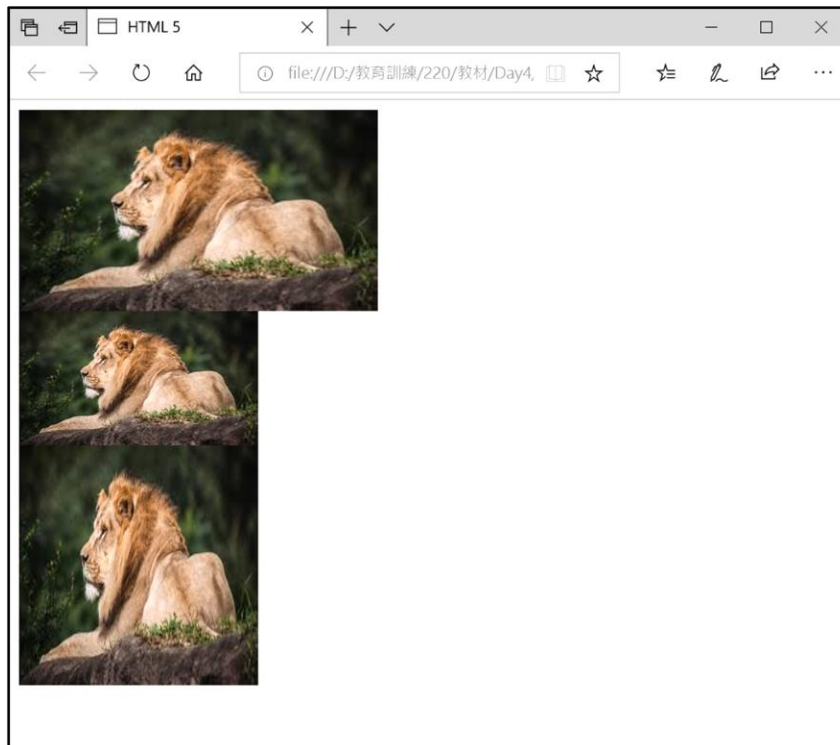
#### 解答：

```
<html>
  <head>
    <title>HTML 4</title>
  </head>
  <body>
    <p align='left'>
```

```
Animals are multicellular, eukaryotic organisms of the kingdom
Animalia (also called Metazoa). All animals are motile, meaning
Animals are multicellular, eukaryotic organisms of the kingdom
Animalia (also called Metazoa). All animals are motile, meaning
</p>
<p align='right'>
Animals are multicellular, eukaryotic organisms of the kingdom
Animalia (also called Metazoa). All animals are motile, meaning
Animals are multicellular, eukaryotic organisms of the kingdom
Animalia (also called Metazoa). All animals are motile, meaning
</p>
<p align='center'>
Animals are multicellular, eukaryotic organisms of the kingdom
Animalia (also called Metazoa). All animals are motile, meaning
Animals are multicellular, eukaryotic organisms of the kingdom
Animalia (also called Metazoa). All animals are motile, meaning
</p>
<p align='justify'>
Animals are multicellular, eukaryotic organisms of the kingdom
Animalia (also called Metazoa). All animals are motile, meaning
Animals are multicellular, eukaryotic organisms of the kingdom
Animalia (also called Metazoa). All animals are motile, meaning
</p>
<body>
</html>
```

## 問題 5：

呈現圖片

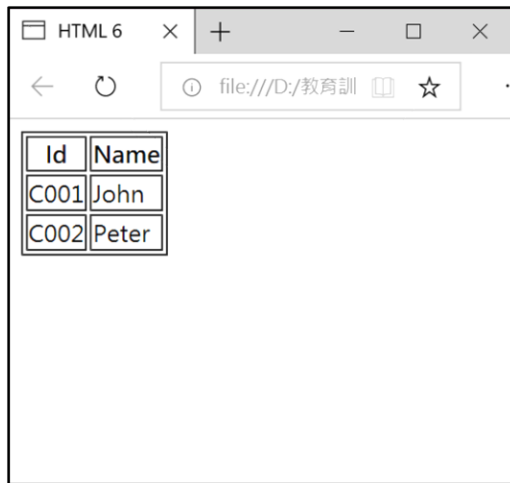


解答：

```
<html>
  <head>
    <title>HTML 5</title>
  </head>
  <body>
    
    <br />
    
    <br />
    
    <br />
  </body>
</html>
```

**問題 6：**

表格的呈現



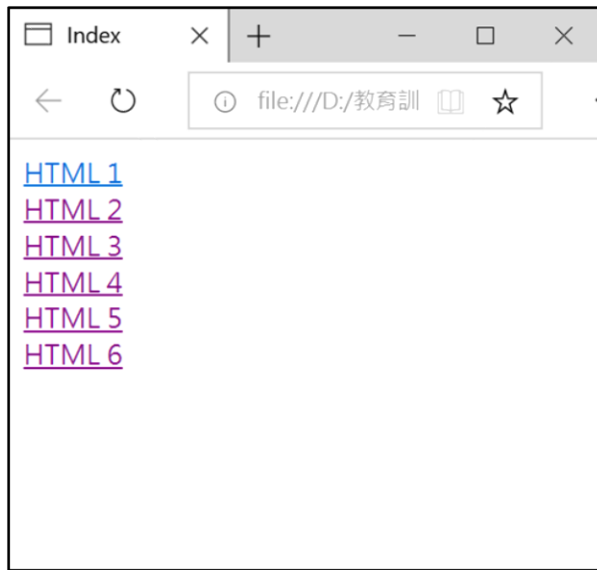
| Id   | Name  |
|------|-------|
| C001 | John  |
| C002 | Peter |

**解答：**

```
<html>
<head>
  <title>HTML 6</title>
</head>
<body>
  <table border="1">
    <tr>
      <th>Id</th>
      <th>Name</th>
    </tr>
    <tr>
      <td>C001</td>
      <td>John</td>
    </tr>
    <tr>
      <td>C002</td>
      <td>Peter</td>
    </tr>
  </table>
</body>
</html>
```

**問題 7：**

超連結

**解答：**

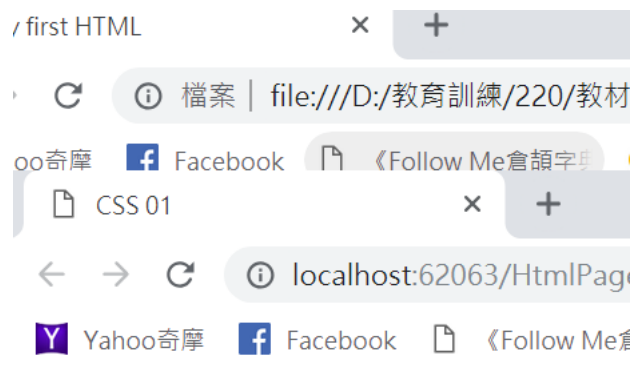
```
<html>
<head>
  <title>Index</title>
</head>
<body>
  <a href="h1.html">HTML 1</a><br />
  <a href="h2.html">HTML 2</a><br />
  <a href="h3.html">HTML 3</a><br />
  <a href="h4.html">HTML 4</a><br />
  <a href="h5.html">HTML 5</a><br />
  <a href="h6.html">HTML 6</a><br />
</body>
</html>
```



## CSS

### 問題 1：

in-line style



Animals are multicellular, eukaryotic organisms of the kingdom Animalia (also called Metazoa). All animals are motile, meaning they can move spontaneously and independently, at some point in their lives. Their body plan eventually becomes fixed as they develop, although some undergo a process of metamorphosis later on in their lives. All animals are **heterotrophs**: they must ingest other organisms or their products for sustenance.

Animals are multicellular, eukaryotic organisms of the kingdom Animalia (also called **Metazoa**). All animals are motile, meaning they can move spontaneously and independently, at some point in their lives. Their body plan eventually becomes fixed as they develop, although some undergo a process of metamorphosis later on in their lives. All animals are **heterotrophs**: they must ingest other organisms or their products for sustenance.

利用 inline-style 的 CSS 調整以上的文字風格

解答：

```
<html>
<!DOCTYPE html>
<html>
<head>
  <title>CSS 01</title>
</head>
<body>
  <div style="border:2px solid black;
    height:300px; width:300px;
    margin:20px; padding:10px;
    color:blue;background-color:lightyellow;
    font-family:Verdana; font-size:16px;">
    Animals are multicellular, eukaryotic organisms of the kingdom Animalia (also
    called Metazoa). All animals are motile, meaning they can move spontaneously and
    independently, at some point in their lives. Their body plan eventually becomes
    fixed as they develop, although some undergo a process of metamorphosis later on
    in their lives. All animals are <span style="color:yellow;background-color:
    red;" > heterotrophs</span>: they must ingest other organisms or their products
    for sustenance.
  </div>
  <hr />
  <div style="border:2px solid black;
    height:300px; width:300px;
    margin:20px; padding:10px;
    color:blue;background-color:lightyellow;
    font-family:Verdana; font-size:16px;">
    Animals are multicellular, eukaryotic organisms of the kingdom Animalia (also
    called Metazoa). All animals are motile, meaning they can move spontaneously and
    independently, at some point in their lives. Their body plan eventually becomes
    fixed as they develop, although some undergo a process of metamorphosis later on
    in their lives. All animals are <span style="color:yellow;background-
    color:red;"> heterotrophs</span>: they must ingest other organisms or their
    products for sustenance.
  </div>
</body>
</html>
```

**問題 2：**

將 style 寫在<head>中(Embedded 方式)

**解答：**

```
<!DOCTYPE html>
<html>
<head>
  <title></title>
  <meta charset="utf-8" />
  <style>
    div {
      border: 2px solid black;
      height: 300px;
      width: 300px;
      margin: 20px;
      padding: 10px;
      color: blue;
      background-color: lightyellow;
      font-family: Verdana;
      font-size: 16px;
    }

    span {
      color: yellow;
      background-color: red;
    }
  </style>
</head>
<body>
  <div>
    Animals are multicellular, eukaryotic organisms of the kingdom Animalia
    (also called Metazoa). All animals are motile, meaning they can move
    spontaneously and independently, at some point in their lives. Their body plan
    eventually becomes fixed as they develop, although some undergo a process of
    metamorphosis later on in their lives. All animals are
    <span>heterotrophs</span>: they must ingest other organisms or their products
    for sustenance.
```

```
</div>
<hr />
<div>
    Animals are multicellular, eukaryotic organisms of the kingdom Animalia
    (also called Metazoa). All animals are motile, meaning they can move
    spontaneously and independently, at some point in their lives. Their body plan
    eventually becomes fixed as they develop, although some undergo a process of
    metamorphosis later on in their lives. All animals are
    <span>heterotrophs</span>: they must ingest other organisms or their products
    for sustenance. </div>
</body>
</html>
```

### 問題 3：

Animals are multicellular, eukaryotic organisms of the kingdom Animalia (also called Metazoa). All animals are motile, meaning they can move spontaneously and independently, at some point in their lives. Their body plan eventually becomes fixed as they develop, although some undergo a process of metamorphosis later on in their lives. All animals are heterotrophs: they must ingest other organisms or their products for sustenance.

Animals are multicellular, eukaryotic organisms of the kingdom Animalia (also called Metazoa). All animals are motile, meaning they can move spontaneously and independently, at some point in their lives. Their body plan eventually becomes fixed as they develop, although some undergo a process of metamorphosis later on in their lives. All animals are heterotrophs: they must ingest other organisms or their products for sustenance.

利用 class, id 設定不同樣式

解答：

```
<!DOCTYPE html>
<html>
<head>
  <title></title>
  <meta charset="utf-8" />
  <style>
    div {
      border: 2px solid black;
      height: 300px;
      width: 300px;
      margin: 20px;
      padding: 10px;
      font-family: Verdana;
      font-size: 16px;
    }

    #div01 {
      color: blue;
      background-color: lightyellow;
    }

    #div02 {
      color: green;
      background-color: pink;
    }

    .m1 {
      color:white;
      background-color:blue;
    }
    .m2 {
      color:white;
      background-color:red;
    }
  </style>
</head>
<body>
```

```
<div id="div01">
    Animals are multicellular, eukaryotic organisms of the kingdom Animalia
    (also called Metazoa). All animals are motile, meaning they can move
    spontaneously and independently, at some point in their lives. Their body plan
    eventually becomes fixed as they develop, although some undergo a process of
    <span class="m1">metamorphosis</span> later on in their lives. All animals are
    <span class="m1">heterotrophs</span>: they must ingest other organisms or their
    products for sustenance. </div>

    <hr />
    <div id="div02">
        Animals are multicellular, eukaryotic organisms of the kingdom Animalia
        (also called <span class="m2">Metazoa</span> All animals are motile, meaning
        they can move spontaneously and independently, at some point in their lives.
        Their body plan eventually becomes fixed as they develop, although some undergo
        a process of metamorphosis later on in their lives. All animals are <span
        class="m2">heterotrophs</span>: they must ingest other organisms or their
        products for sustenance. </div>
    </body>
</html>
```

#### 問題 4：

CSS 獨立成為檔案

#### 解答：

htmlpage.html

```
<!DOCTYPE html>
<html>
<head>
    <title></title>
    <meta charset="utf-8" />
    <link href="style/Site.css" rel="stylesheet" />
</head>
<body>
    <div id="div01">
        Animals are multicellular, eukaryotic organisms of the kingdom Animalia
        (also called Metazoa). All animals are motile, meaning they can move
        spontaneously and independently, at some point in their lives. Their body plan
```

```
eventually becomes fixed as they develop, although some undergo a process of
<span class="m1"> metamorphosis</span> later on in their lives. All animals are
<span class="m1"> heterotrophs</span>: they must ingest other organisms or their
products for sustenance. </div>

<hr />

<div id="div02">
    Animals are multicellular, eukaryotic organisms of the kingdom Animalia
    (also called <span class="m2">Metazoa</span> All animals are motile, meaning
    they can move spontaneously and independently, at some point in their lives.
    Their body plan eventually becomes fixed as they develop, although some undergo
    a process of metamorphosis later on in their lives. All animals are <span
    class="m2"> heterotrophs</span>: they must ingest other organisms or their
    products for sustenance. </div>
</body>
</html>
```

style/style.css

```
div {
    border: 2px solid black;
    height: 300px;
    width: 300px;
    margin: 20px;
    padding: 10px;
    font-family: Verdana;
    font-size: 16px;
}

#div01 {
    color: blue;
    background-color: lightyellow;
}

#div02 {
    color: green;
    background-color: pink;
}

.m1 {
```

```
    color: white;
    background-color: blue;
}

.m2 {
    color: white;
    background-color: red;
}
```

## JavaScript

### 問題 1：

跳出訊息視窗”Hello world!”

解答：

```
<!DOCTYPE html>
<html>
<head>
    <title></title>
    <meta charset="utf-8" />
    <script>
        function btnOk_Click() {
            alert("Hello world!");
        }
    </script>
</head>
<body>
    <input type="button" value="OK" onclick="btnOk_Click();" />
</body>
</html>
```

### 問題 2：

視窗中配置按鈕(<input type="button">)、標籤(<label>)、文字方塊(<input>)，按下按鈕後，在文字方塊中出現 Hello world。

解答：

```
<!DOCTYPE html>
<html>
<head>
```



```
<title></title>
<meta charset="utf-8" />
<script>
    function btnOk_Click() {
        var txt = document.getElementById("txtData");
        txt.value = "Hello world!";
    }
</script>
</head>
<body>
    data:<input type="text" id="txtData" />
    <input type="button" onclick="btnOk_Click();" value="OK" />
</body>
</html>
```

**問題 3：**

文字方塊 A, B 中輸入數值，計算數值相加、減、乘、除的結果。

**解答：**

```
<!DOCTYPE html>
<html>
<head>
    <title></title>
    <meta charset="utf-8" />
    <script>
        function btnAdd_Click() {
            var a = parseFloat(document.getElementById("txtA").value);
            var b = parseFloat(document.getElementById("txtB").value);
            document.getElementById("txtAns").value = a + b;
        }

        function btnSub_Click() {
            var a = parseFloat(document.getElementById("txtA").value);
            var b = parseFloat(document.getElementById("txtB").value);
            document.getElementById("txtAns").value = a - b;
        }

        function btnMul_Click() {
```

```
        var a = parseFloat(document.getElementById("txtA").value);
        var b = parseFloat(document.getElementById("txtB").value);
        document.getElementById("txtAns").value = a * b;
    }

    function btnDiv_Click() {
        var a = parseFloat(document.getElementById("txtA").value);
        var b = parseFloat(document.getElementById("txtB").value);
        document.getElementById("txtAns").value = a / b;
    }
</script>
</head>
<body>
    A=<input id="txtA" type="text" /><br />
    B=<input id="txtB" type="text" /><br />
    Ans=<input id="txtAns" type="text" /><br />
    <input id="btnAdd" type="button" value="+" onclick="btnAdd_Click();" />
    <input id="btnSub" type="button" value="-" onclick="btnSub_Click();" />
    <input id="btnMul" type="button" value="*" onclick="btnMul_Click();" />
    <input id="btnDiv" type="button" value="/" onclick="btnDiv_Click();" />
</body>
</html>
```

**問題 4：**

輸入成績，判斷是否及格(低於 60 不及格，文字方塊為黃底紅字，高於或等於 60 及格，文字方塊為藍底白字)。

**解答：**

```
<!DOCTYPE html>
<html>
<head>
    <title></title>
    <meta charset="utf-8" />
    <script>
        function btnOk_Click() {
            var score = parseInt(document.getElementById("txtScore").value, 10);
            var txt = document.getElementById("txtResult");
            if (isNaN(score)) {
```

```

        txt.value = "";
        txt.style.color = "black";
        txt.style.backgroundColor = "white";
        alert("Score 必須是整數");
    }
    else if (score < 60) {
        txt.value = "不及格";
        txt.style.color = "red";
        txt.style.backgroundColor = "yellow";
    } else if (score >= 60) {
        txt.value = "及格";
        txt.style.color = "white";
        txt.style.backgroundColor = "blue";
    }
}
</script>
</head>
<body>
    Score:<input id="txtScore" type="text" /><br />
    Result:<input id="txtResult" type="text" /><br />
    <input type="button" value="OK" onclick="btnOk_Click();" />
</body>
</html>

```

**問題 5：**

輸入幣別(U=USD/J=JPY/E=EUR)，台幣金額，按照匯率

NT:USD = 1: 30.05，NT:JPY = 1:0.2785，NT:EUR = 1:32.55

(可依現在匯率調整數字)

**解答：**

```

<!DOCTYPE html>
<html>
<head>
    <title></title>
    <meta charset="utf-8" />
    <script>
        function btnOk_Click() {

```

```
        var nt = parseInt(document.getElementById("txtNT").value, 10);
        var curr = parseFloat(document.getElementById("selCurr").value);
        document.getElementById("txtAmount").value = nt / curr;
    }
</script>
</head>
<body>
    NT=<input id="txtNT" type="text" /><br />
    幣別=<select id="selCurr">
        <option value="30.05">美金</option>
        <option value="0.2785">日圓</option>
        <option value="32.55">歐元</option>
    </select><br />
    可兌換金額<input id="txtAmount" type="text" /><br />
    <input type="button" onclick="btnOk_Click();" value="兌換" />
</body>
</html>
```

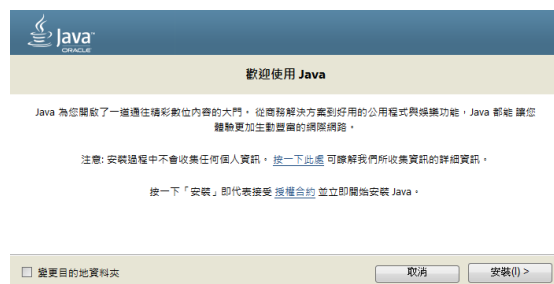
# Part-7 其它程式語言 Java, Python

## 問題 1.

### 下載並安裝 Java Runtime Environment(JRE)

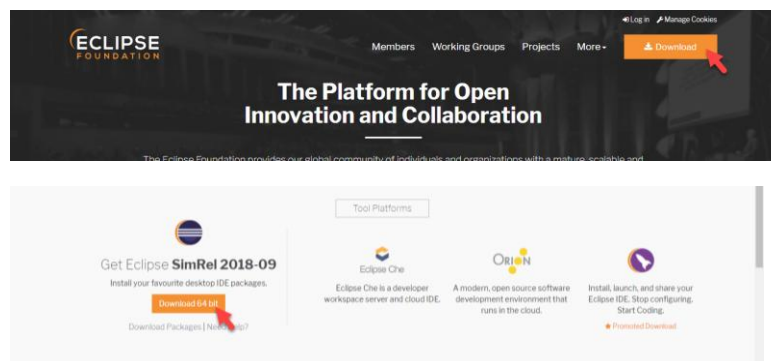


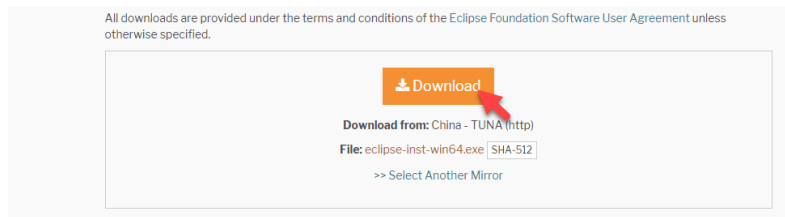
### 尋找 Java 官方網站並下載



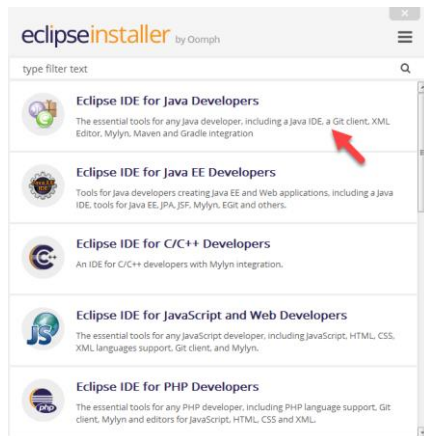
### 開始安裝程式

### 下載並安裝 Eclipse

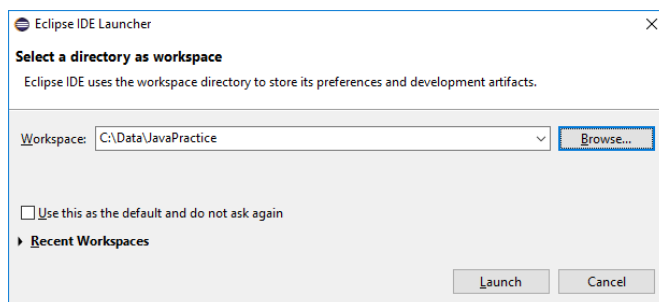




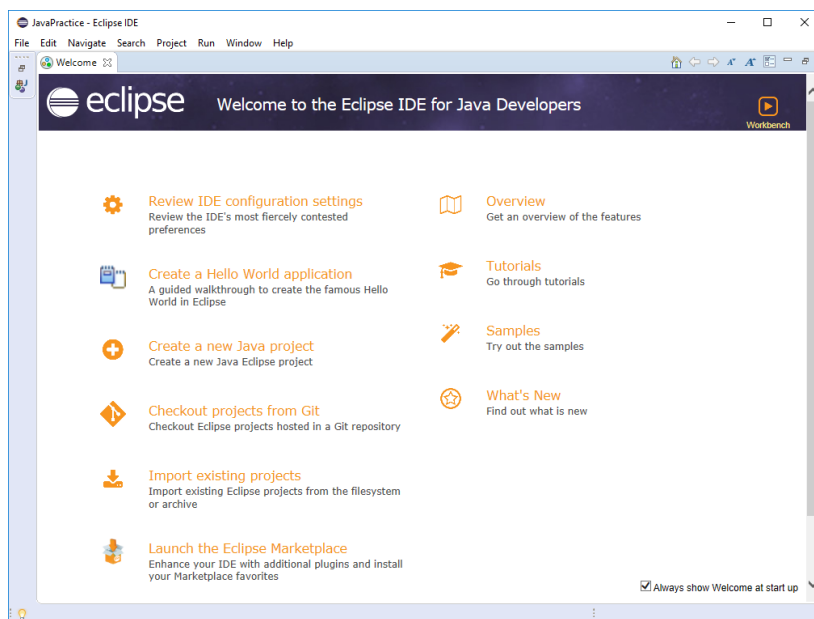
尋找 Eclipse 官方網站點選下載



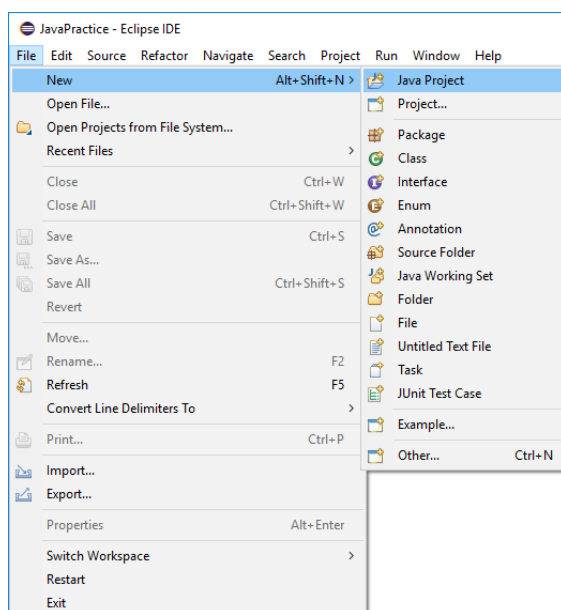
安裝程式時，依據需求選擇選項安裝



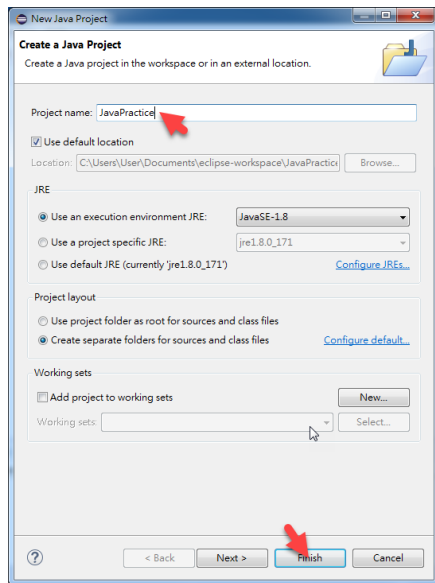
使用 Eclipse 建立 Java 專案，撰寫第一個 Java 程式，在畫面上顯示 Hello world。



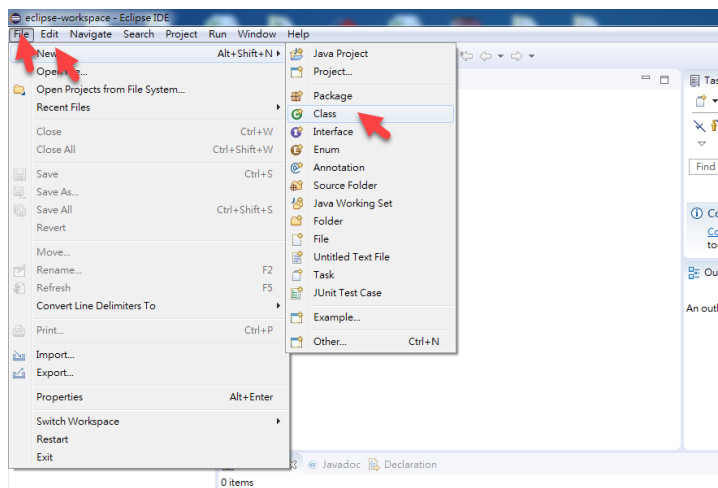
第一次啟動的時候會要求設定工作區



點選 File>New>Java Project 建立專案

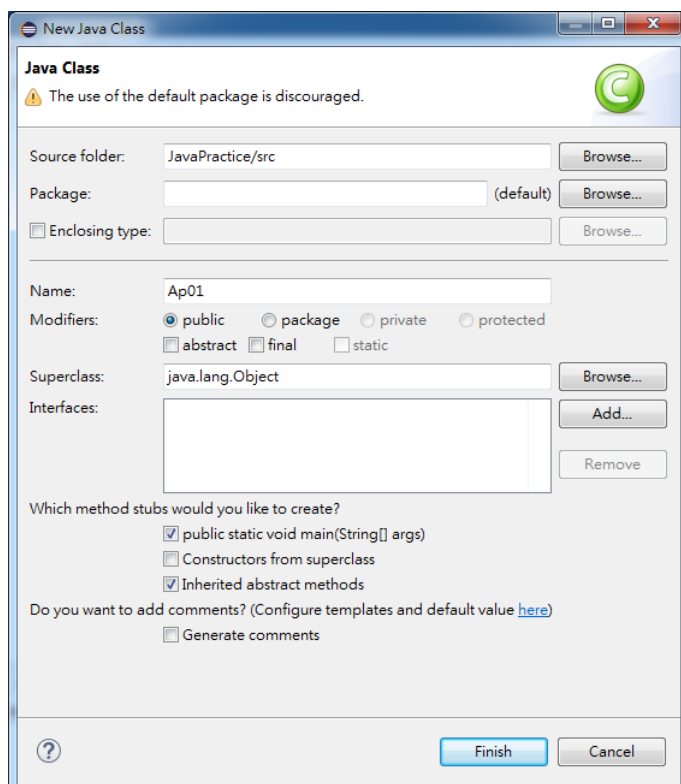


輸入專案名稱並點選 **Finish** 完成



點選 **File>New>Class** 建立類別



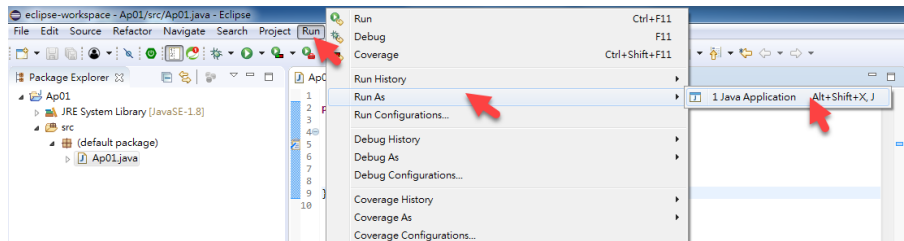


輸入類別稱，並勾選 `public static void main(String[] args)` 來自動建立 `main` 方法，並且點選 `Finish` 完成

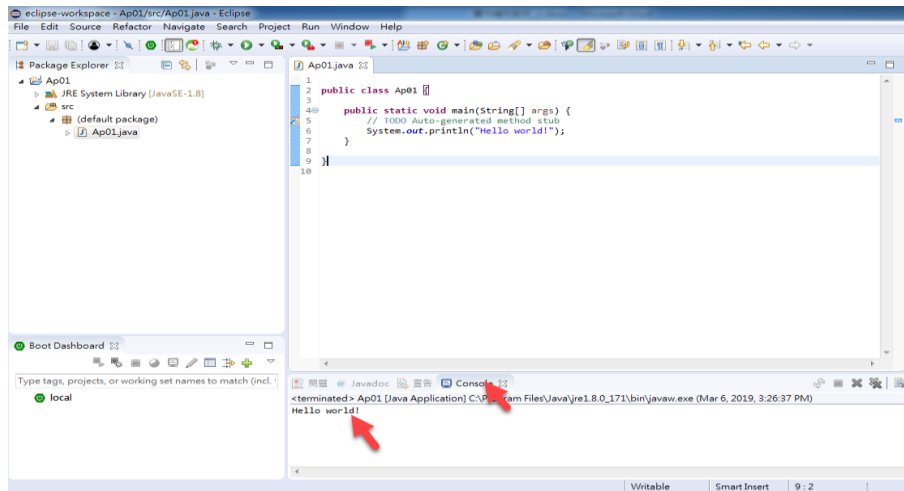
```
public class Ap01 {  
    public static void main(String[] args) {  
        // TODO Auto-generated method stub  
        System.out.println("Hello world!");  
    }  
}
```

```
public class Ap01 {  
    public static void main(String[] args) {  
        // TODO Auto-generated method stub  
        System.out.println("Hello world!");  
    }  
}
```

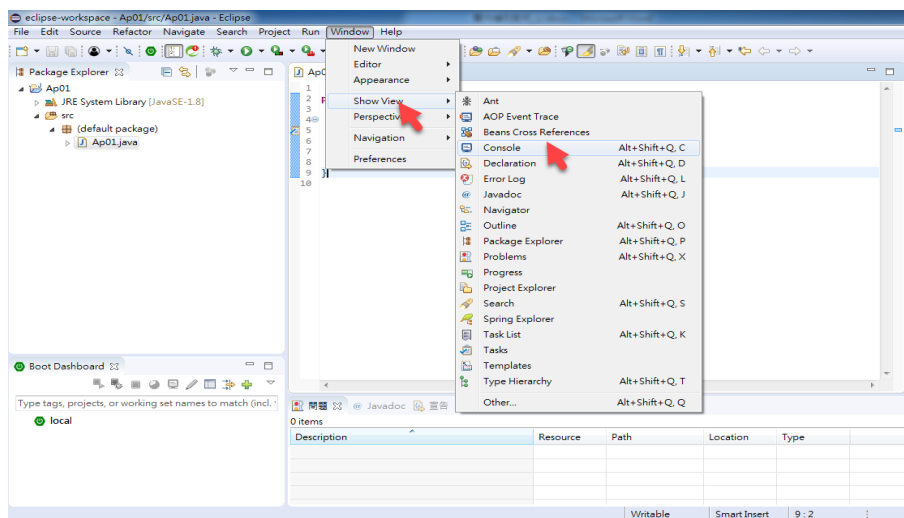
在 `main` 方法中輸入程式碼



點選 Run>Run As>1.Java Application



可在下方 Console 中看到結果



如果沒有 Console 視窗的話，點選 Window>Show View>Console

**問題 2.**

使用畫面上輸入姓名(name)，顯示”Hello, (輸入姓名)!”。

```
import java.util.Scanner;

public class Ap02 {

    public static void main(String[] args) {
        String s;
        Scanner sc = new Scanner(System.in);

        System.out.print("Your name:");
        s = sc.nextLine();
        sc.close();
        System.out.printf("Hello, %s", s);
    }
}
```

**問題 1.**

宣告變數 **a**, **b** 分別為 **short a**; **int b**; 指定 **b = 100000** 再將 **b** 的值指定給 **a**, 顯示 **a**, **b** 的結果(十進位制)以及 16 進位碼。

```
public class Ap01 {  
  
    public static void main(String[] args) {  
        short a;  
        int b;  
        b = 100000;  
        a = (short)b;  
        System.out.printf("a=%d, b=%d", a, b);  
    }  
}
```

**問題 2.**

宣告變數 **n** 為整數, **int n**; 以及 **s** 為字串, **String s**; 設定 **n = 1234**; 並將 **n** 的結果給 **s**。

```
public class Ap02 {  
  
    public static void main(String[] args) {  
        int n;  
        String s;  
        n = 1234;  
        s = n.toString();  
        System.out.printf("n=%d, s=%s", n, s);  
    }  
}
```

**問題 3.**

輸入 **a**, **b** 兩個數字, 將兩數相加結果顯示在畫面上, 完成偽碼(Pseudocode)以及 Java 程式碼, 並執行測試結果。

解答：

**Pseudocode**

輸入 **a**

輸入 **b**

計算 **a + b** 的結果

程式碼

```
import java.util.Scanner;

public class Ap03 {

    public static void main(String[] args) {
        String s;
        int a, b;
        Scanner sc = new Scanner(System.in);

        //輸入 a
        System.out.print("a=");
        a = sc.nextInt();

        //輸入 b
        System.out.print("b=");
        b = sc.nextInt();
        sc.close();

        //計算 a + b 的結果
        System.out.printf("%d + %d = %d", a, b, a + b);
    }
}
```

**問題 4.**

輸入成績，判斷是否及格(低於 60 不及格，高於或等於 60 及格)，完成偽碼(Pseudocode)以及 Java 程式碼，並執行測試結果。

解答：

**Pseudocode**

輸入成績(score)

IF score < 60 THEN

    顯示 “不及格”

ELSE

    顯示 “及格”

END IF

## 程式碼

```
import java.util.Scanner;

public class Ap04 {

    public static void main(String[] args) {
        int score;
        Scanner sc = new Scanner(System.in);

        //輸入成績(score)
        System.out.print("成績:");
        score = sc.nextInt();
        sc.close();

        if(score < 60)
            System.out.println("不及格");
        else
            System.out.println("及格");
    }
}
```

## 問題 5.

輸入成績，判斷等級，低於 60(不含等於 60)，顯示 “C-Class”，  
60 以上(含等於 60)，80 以下(不含等於 80)，顯示 “B-Class”，  
80 以上(包含等於 80)，顯示 A-Class。

完成偽碼(Pseudocode)以及 Java 程式碼，並執行測試結果。

解答：

## Pseudocode

```
輸入成績(score)
IF score < 60 THEN
    顯示 “C-Class”
ELSEIF score < 80 THEN
    顯示 “B-Class”
ELSE
    顯示 “A-Class”
END IF
```

## 程式碼

```

import java.util.Scanner;

public class Ap05 {

    public static void main(String[] args) {
        int score;
        Scanner sc = new Scanner(System.in);

        //輸入成績(score)
        System.out.print("成績:");
        score = sc.nextInt();
        sc.close();

        if (score < 60)
            System.out.println("C-Class");
        else if(score < 80)
            System.out.println("B-Class");
        else
            System.out.println("A-Class");
    }
}

```

## 問題 6.

所得稅級距公式如下

107 年度綜合所得稅速算公式一覽表（單位：新台幣元）

| 綜合所得淨額    |   |           |   | 稅率  |   | 累進差額    |   | 應納稅額 |
|-----------|---|-----------|---|-----|---|---------|---|------|
| 0         | ~ | 540,000   | × | 5%  | - | 0       | = |      |
| 540,001   | ~ | 1,210,000 | × | 12% | - | 37,800  | = |      |
| 1,210,001 | ~ | 2,420,000 | × | 20% | - | 134,600 | = |      |
| 2,420,001 | ~ | 4,530,000 | × | 30% | - | 376,600 | = |      |
| 4,530,001 | ~ | 以上        | × | 40% | - | 829,600 | = |      |

輸入淨所得額，計算應納稅額，完成流程圖(Flowchart)、偽碼(Pseudocode)以及 Java 程式碼，並執行測試結果。

解答：

#### Pseudocode

```
輸入年所得收入(income)
IF income <= 540000 THEN
    所得稅(tax) = income * 0.05
ELSEIF income <= 1210000 THEN
    tax = income * 0.12 - 37800
ELSEIF income <= 2420000 THEN
    tax = income * 0.2 - 134600
ELSEIF income <= 4530000 THEN
    tax = income * 0.3 - 376600
ELSE
    tax = income * 0.4 - 899600
END IF
顯示 年所得收入(income)，應繳所得稅(tax)
```

#### 程式碼

```
import java.util.Scanner;

public class Ap06 {

    public static void main(String[] args) {
        int income, tax;
        Scanner sc = new Scanner(System.in);

        //輸入年所得收入(income)
        System.out.print("年所得收入:");
        income = sc.nextInt();
        sc.close();

        if (income <= 540000)
            tax = (int)(income * 0.05 + 0.5);
        else if (income <= 1210000)
            tax = (int)(income * 0.12 - 37800 + 0.5);
        else if (income <= 2420000)
            tax = (int)(income * 0.2 - 134600 + 0.5);
        else if (income <= 4530000)
            tax = (int)(income * 0.3 - 376600 + 0.5);
```



```
        else
            tax = (int)(income * 0.4 - 899600 + 0.5);

            //顯示 年所得收入(income), 應繳所得稅(tax)
            System.out.printf("年所得收入:%d, 應納所得稅:%d", income, tax);
        }
    }
}
```

### 問題 7.

輸入月份，

1, 3, 5, 7, 8, 10, 12 月顯示 大月(31 天)；

4, 6, 9, 11 月顯示 小月(30 天)；

2 月顯示 二月(28 或 29 天)。

完成偽碼(Pseudocode)以及 Java 程式碼，並執行測試結果。

解答：

#### Pseudocode

輸入月份(mn)

CASE mn

1, 3, 5, 7, 8, 10, 12:

顯示 “大月(31 天)”

4, 6, 9, 11:

顯示 “小月(30 天)”

2:

顯示 “二月(28 或 29 天)”

ELSE

顯示 “僅接受 1-12”

END CASE

#### 原始程式碼

```
import java.util.Scanner;

public class Ap07 {

    public static void main(String[] args) {
        int mn;
        Scanner sc = new Scanner(System.in);
```

```
//輸入月份(mn)
System.out.print("月份:");
mn = sc.nextInt();
sc.close();

/*
CASE mn
1, 3, 5, 7, 8, 10, 12:
    顯示 "大月(31 天)"
4, 6, 9, 11:
    顯示 "小月(30 天)"
2:
    顯示 "二月(28 或 29 天)"
ELSE
    顯示 "僅接受 1 - 12"
END CASE
*/
switch(mn)
{
    case 1:
    case 3:
    case 5:
    case 7:
    case 8:
    case 10:
    case 12:
        System.out.println("大月(31 天)");
        break;
    case 4:
    case 6:
    case 9:
    case 11:
        System.out.println("小月(30 天)");
        break;
    case 2:
        System.out.println("二月(28 或 29 天)");
        break;
    default:
        System.out.println("僅接受 1 - 12");
}
```

```
                break;
            }
        }
    }
```

**問題 8.**

輸入幣別(U=USD/J=JPY/E=EUR)，台幣金額，按照匯率

NT:USD = 1: 30.05，NT:JPY = 1:0.2785，NT:EUR = 1:32.55(可依現在匯率調整數字)

計算可兌換外幣。完成流程圖(Flowchart)、偽碼(Pseudocode)以及 Java 程式碼，並執行測試結果。

解答：

**Pseudocode**

輸入台幣 nt

輸入幣別 curr(U=USD/J=JPY/E=EUR)

CASE curr

“U”:

amt = nt / 30.05

“J”:

amt = nt / 0.2758

“E”:

amt = nt / 32.55

ELSE

amt = 0

END CASE

顯示台幣金額(nt)、可兌換外幣(amt)

**程式碼**

```
import java.util.Scanner;

public class Ap08 {

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        String curr;
        int nt;
```

```
double amt;

//輸入台幣nt
System.out.print("台幣:");
nt = sc.nextInt();

//輸入幣別curr(U = USD / J = JPY / E = EUR)
System.out.print("幣別(U=USD/J=JPY/E=EUR):");
curr = sc.nextLine();
sc.close();
switch (curr)
{
    case "U":
        amt = nt / 30.5;
        break;
    case "J":
        amt = nt / 0.2758;
        break;
    case "E":
        amt = nt / 32.55;
        break;
    default:
        amt = 0;
        break;
}

//顯示台幣金額(nt)、可兌換外幣(amt)
System.out.printf("台幣:%d, 可兌換外幣:%f", nt, amt);
}

}
```

**問題 9.**

在畫面上顯示 0 到 9

解答：

程式碼

```
public class Ap09 {
    public static void main(String[] args) {
        for(int i = 0; i<10; i++)
            System.out.println(i);
    }
}
```

**問題 10.**

畫星星，輸入數字在畫面上以 \* 印出下圖

完成偽碼(Pseudocode)以及 Java 程式碼，並執行測試結果。

1.

```
*
**
***
...
**** ..... *
(n 個*)
```

2.

```
**** ..... *(n 個*)
****.....*(n-1 個*)
....
***
**
*
```

3.

```
      *
    ***
  *****
...
***** (2 * n - 1 個 *)
```

解答：

1.

Pseudocode

```
輸入 n
FOR i=1 TO n
    畫 i 個 "*"
    跳行
END FOR
```

程式碼

```
import java.util.Scanner;

public class Ap10_1 {

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n;
        //輸入 n
        System.out.print("n=");
        n = sc.nextInt();
        sc.close();

        for(int i = 1; i<=n; i++)
        {
            //畫 i 個 "*"
            for(int j= 1; j<=i; j++)
                System.out.print("*");
            //跳行
            System.out.println();
        }
    }
}
```

2.

**Pseudocode**

```
輸入 n
FOR i=n TO 1
    畫 i 個 “*”
    跳行
END FOR
```

**程式碼**

```
import java.util.Scanner;

public class Ap10_2 {

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n;
        //輸入 n
        System.out.print("n=");
        n = sc.nextInt(s);
        sc.close();

        for (int i = n; i >= 1; i--)
        {
            //畫 i 個 “*”
            for(int j= 1; j<=i; j++)
                System.out.print("*");
            //跳行
            System.out.println();
        }
    }
}
```

3.

**Pseudocode**

```
輸入 n
FOR i=n TO 1
    畫 n - i 個 空白
    畫 2 * i - 1 個 “*”
```

跳行

END FOR

程式碼

```
import java.util.Scanner;

public class Ap10_3 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n;
        //輸入 n
        System.out.print("n=");
        n = sc.nextInt();
        sc.close();

        for (int i = 1; i <= n; i++)
        {
            //畫 n - i 個空白
            for (int j = 1; j <= n - i; j++)
                System.out.print(" ");
            //畫 2 * i - 1 個 "*"
            for (int j = 1; j <= 2 * i - 1; j++)
                System.out.print("*");
            //跳行
            System.out.println();
        }
    }
}
```

#### 問題 11.

輸入本金、利率(%)、期數，列出每一期的期數、本金、利息、本利和。

完成流偽碼(Pseudocode)以及 Java 程式碼，並執行測試結果。



解答：

**Pseudocode**

```
輸入本金(principal)
輸入利率(rate:%)
期數(n)

FOR i = 1 TO n
    利息(interest) = principal * rate%
    顯示 期數(i)，本金(principal)，利息(interest)，本金+利息
    將利息累加至本金
END FOR
```

**程式碼**

```
import java.util.Scanner;

public class Ap11 {

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int principal, n;
        double rate;

        // 輸入本金
        System.out.print("本金=");
        principal = sc.nextInt();

        // 輸入利率(%)
        System.out.print("利率(%)=");
        rate = sc.nextDouble();

        // 輸入期數
        System.out.print("期數=");
        n = sc.nextInt();
        sc.close();
        System.out.println("期 數    本        金        利        息        合
計");
        System.out.println("====  =====  =====  =====");
        for(int i = 1; i<=n; i++)
```

```

    {
        int interest = (int)(principal * rate / 100 + 0.5);
        System.out.printf("% 4d % 10d % 10d % 10d\n",
            i, principal, interest, principal + interest);
        principal += interest;
    }
    System.out.println("==== ===== ===== =====");
}
}
}

```

**問題 12.**

完成 九九乘法表

完成偽碼(Pseudocode)以及 Java 程式碼，並執行測試結果。

解答：

**Pseudocode**

```

FOR i = 1 TO 9
    顯示 i * 1 = i 到 i * 9 = i
    (i=1, 顯示: 1 * 1 = 1  1 * 2 = 2  ....  1 * 9 = 9)
    (i=2, 顯示: 2 * 1 = 2  2 * 2 = 4  ....  2 * 9 = 18)
    .....
    (i=9, 顯示: 9 * 1 = 9  9 * 2 = 18  ....  9 * 9 = 81)
END FOR

```

**程式碼**

```

public class Ap12 {

    public static void main(String[] args) {
        for(int i =1; i<=9; i++)
        {
            for (int j = 1; j <= 9; j++)
                System.out.printf("%d*%d=%2d  ", j, i, i * j);
            System.out.println();
        }
    }
}

```

**問題 13.**

使用者輸入數值，直到輸入 0 為止，再將所輸入的數值加總的結果顯示在畫面上。完成偽碼 (Pseudocode) 以及 Java 程式碼，並執行測試結果。

解答：

**Pseudocode**

**REPEAT**

    輸入整數 *n*

    將 *n* 累加至 *total*

**UNTIL** *n* = 0

**程式碼**

```
import java.util.Scanner;

public class Ap13 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n, total = 0;
        do
        {
            //輸入整數 n
            System.out.print("n(0 for end)=");
            n = sc.nextInt();

            //將n 累加至 total
            total += n;
        } while (n != 0);
        sc.close();
        System.out.printf("小計:%d ", total);
    }
}
```

**問題 14**

隨機產生 1 到 99 的亂數，作為答案，讓使用者猜數字，直到猜對為止

**解答****Pseudocode**

產生 1 - 99 的亂數 ans

REPEAT

    使用者猜數字 guess

    IF guess < ans THEN

        顯示 "高一點"

    ELSE IF guess > ans THEN

        顯示 "低一點"

    ELSE

        顯示 "猜對了"

    ENDIF

UNTIL 猜對了

**程式碼**

```
import java.util.Scanner;
import java.util.Random;

public class Ap14 {

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        Random rnd = new Random();
        int ans, guess;

        //產生 1 - 99 的亂數 ans, Random.nextInt(n) 會產生 0~n-1 之間的數
        ans = rnd.nextInt(99)+1;
        sc.close();

        do
        {
            System.out.print("猜數字(1-99):");
            guess = sc.nextInt();

            if(guess < ans)
```

```
        System.out.println("高一點");
    else if(guess > ans)
        System.out.println("低一點");
    else
        System.out.println("猜對了");
    } while (ans!=guess);
}
}
```

### 問題 15.

隨機產生四個 1~6 的數字，利用以下的骰子計算方式顯示點數結果

1. 四個骰子的點數相同，顯示 “一色”。
2. 三個骰子的點數相同，另一個骰子點數不同，顯示”沒點，重擲”。
3. 兩個骰子的點數相同，另外兩個骰子的點數不相同，顯示不相同的兩個骰子的點數合計。
4. 兩個骰子的點數相同，另外兩個骰子的點數也相同，取較大的兩個相同點數的合計為點數。
5. 四個骰子的點數都不同，顯示”沒點，重擲”。

完成偽碼(Pseudocode)以及 Java 程式碼，並執行測試結果。

### 解答

#### Pseudocode

產生四個 1 - 6 的亂數

IF 四個骰子的點數相同 THEN

顯示 “一色”

ELSEIF 三個骰子的點數相同，另一個骰子點數不同 THEN

顯示”沒點，重擲”

ELSEIF 兩個骰子的點數相同，另外兩個骰子的點數也相同 THEN

較大的兩個相同點數的合計為點數

ELSEIF 兩個骰子的點數相同，另外兩個骰子的點數不相同 THEN

顯示不相同的兩個骰子的點數合計

ELSE

顯示”沒點，重擲”

END IF

#### 程式碼

```
import java.util.Random;
import java.util.Arrays;
```

```
public class Ap15 {
    public static void main(String[] args) {
        Random rnd = new Random();
        int[] dices = new int[4];

        //產生四個1 - 6 的亂數
        for (int i = 0; i < 4; i++)
        {
            dices[i] = rnd.nextInt(6)+1;
        }
        //將陣列排序
        Arrays.sort(dices);

        //由小至大顯示骰子的點數
        for (int i = 0; i < dices.length; i++)
            System.out.printf("%d ", dices[i]);
        System.out.println();

        if (dices[0] == dices[3])
            System.out.println("一色");
        else if (dices[0] == dices[2] || dices[1] == dices[3])
            System.out.println("沒點重擲");
        else if(dices[0]==dices[1] && dices[2] == dices[3])
            System.out.printf("%d 點%n", dices[2] + dices[3]);
        else if(dices[0] == dices[1])
            System.out.printf("%d 點%n", dices[2] + dices[3]);
        else if(dices[1] == dices[2])
            System.out.printf("%d 點%n", dices[0] + dices[3]);
        else if (dices[2] == dices[3])
            System.out.printf("%d 點%n", dices[0] + dices[1]);
        else
            System.out.println("沒點重擲");
    }
}
```

**問題 16.**

輸入數字  $a$ ,  $b$  計算  $a + b$  的結果，避免使用者因為輸入格式錯誤而中斷程式。

**解答**

```
import java.util.Scanner;

public class Ap16 {

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int a = 0, b = 0;
        String s;
        Boolean incorrect;

        //輸入 a
        do
        {
            System.out.print("a=");
            s = sc.nextLine();
            try {
                a = Integer.parseInt(s);
                incorrect = true;
            } catch (Exception e) {
                incorrect = false;
            }
            if (!incorrect)
                System.out.printf("%s'不是整數，請輸入整數%n", s);
        } while (!incorrect);

        //輸入 b
        do
        {
            System.out.print("b=");
            s = sc.nextLine();
            try {
                b = Integer.parseInt(s);
                incorrect = true;
            } catch (Exception e) {
```

```
        incorrect = false;
    }
    if (!incorrect)
        System.out.printf("%s'不是整數，請輸入整數%n", s);
    } while (!incorrect);
    sc.close();

    //計算 a + b 的結果
    System.out.printf("%d + %d = %d%n", a, b, a + b);
}

}
```

**問題 17.**

寫一個類別 Person

成員變數 private String id, name

成員方法 public void Display() 顯示 id, name

建構函式 public Person(String id, String name) 設定 id, name

解答：

Person.java

```
public class Person {
    public String id, name;
    public void Display()
    {
        System.out.printf("Id:%s, Name:%s%n", id, name);
    }
    public Person(String id, String name)
    {
        this.id = id;
        this.name = name;
    }
}
```



Ap17.java

```
public class Ap17 {  
  
    public static void main(String[] args) {  
        Person p1 = new Person("C001", "John");  
        p1.Display();  
    }  
}
```

**問題 18.**

設計一類別立方體(Cube) ,

成員變數長 private double \_length, \_width, \_height

成員方法體積 public double Volume()

成員方法表面積 public double Area()

建構函式 Cube(double length, double width, double height)

解答：

Cube.java

```
public class Cube {  
    public double _length, _width, _height;  
    public Cube(double length, double width, double height)  
    {  
        _length = length;  
        _width = width;  
        _height = height;  
    }  
  
    public double Volume()  
    {  
        return _length * _width * _height;  
    }  
  
    public double Area()  
    {  
        return 2 * (_length * _width + _length * _height + _width * _height);  
    }  
}
```

Ap18.java

```
public class Ap18 {  
  
    public static void main(String[] args) {  
        Cube c1 = new Cube(6, 6, 6);  
        Cube c2 = new Cube(12, 5, 10);  
        System.out.printf("c1: Area=%f, Volume=%f%n", c1.Area(),  
                           c1.Volume());  
        System.out.printf("c2: Area=%f, Volume=%f%n", c2.Area(),  
                           c2.Volume());  
    }  
}
```

**問題 19.**

設計類別 BMICalculator(BMI 計算器)

成員變數 private double \_height(身高:m), \_width(體重:kg)

成員方法 public double BMI() : 體重(kg) / 身高(m)<sup>2</sup>

成員方法 public string Status():

BMI &lt; 18.5: 體重過輕

18.5 &lt;= BMI &lt; 24: 正常

BMI &gt;= 24: 體重過重

建構函式 public BMICalculator(double weight, double height)

weight: 體重(kg), height: 身高(cm)

解答：

BMICalculator.java

```
public class BMICalculator {  
    private double _height, _weight;  
    private double BMI()  
    {  
        return _weight / _height / _height;  
    }  
  
    public String Status()  
    {  
        double bmivalue = BMI();  
        if (bmivalue < 18.5)  
            return "體重過輕";  
    }  
}
```

```
        else if (bmivalue < 24)
            return "正常";
        else
            return "體重過重";
    }

    public BMICalculator(double weight, double height)
    {
        _weight = weight;
        _height = height / 100;
    }
}
```

Ap19.java

```
public class Ap19 {

    public static void main(String[] args) {
        BMICalculator calc = new BMICalculator(100, 180);
        System.out.println(calc.Status());
    }
}
```

### 問題 20.

設計類別 DiceGame(骰子遊戲)

成員變數 private int[] ans = new int[4]

成員方法 public void Roll() (擲骰子)

成員方法 public int Point() (計算點數)

解答：

Dice.java

```
import java.util.Random;
import java.util.Arrays;

public class DiceGame {
    private int[] dice = new int[4];
    Random rnd;
    public DiceGame()
```

```
{
    rnd = new Random();
}

public void Roll()
{
    for(int i = 0; i<4; i++)
        dice[i] = rnd.nextInt(6)+1;
}

public String Display()
{
    Arrays.sort(dice);

    for (int i = 0; i < dice.length; i++)
        System.out.printf("%d ", dice[i]);

    if(dice[0] == dice[3])
        return "一色";
    else if(dice[0]==dice[2] || dice[1]==dice[3])
        return "沒點重擲";
    else if(dice[0] == dice[1] && dice[2] == dice[3])
        return String.format("%d點", dice[2] + dice[3]);
    else if(dice[0] == dice[1])
        return String.format("%d點", dice[2] + dice[3]);
    else if(dice[1] == dice[2])
        return String.format("%d點", dice[0] + dice[3]);
    else if(dice[2] == dice[3])
        return String.format("%d點", dice[0] + dice[1]);
    else
        return "沒點重擲";
}
}
```

Ap20.java

```
public class Ap20 {

    public static void main(String[] args) {
```

```
DiceGame g1 = new DiceGame();
for (int i = 0; i < 10; i++)
{
    g1.Roll();
    System.out.println(g1.Display());
}
}
```

**問題 21.**

設計類別 GuessGame(猜數字遊戲)

成員變數 private int \_guess, \_ans

成員方法 public void Start() 產生 1-99 亂數(\_ans)

成員方法 public void Guess() 使用者猜 1-99 的數字(\_guess)

成員方法 public bool GotAns() 判斷使用者是否猜對

解答：

GuessGame.java

```
import java.util.Random;
import java.util.Scanner;

public class GuessGame {
    private int _guess, _ans;

    public void Start()
    {
        Random rnd = new Random();
        _ans = rnd.nextInt(99)+1;
    }

    public void Guess()
    {
        Scanner sc = new Scanner(System.in);
        String s;
        System.out.print("猜數字(1-99):");
        _guess = sc.nextInt();
        sc.close();
    }
}
```

```
public Boolean GotAns()  
{  
    if(_guess < _ans)  
        System.out.println("高一點");  
    else if(_guess > _ans)  
        System.out.println("低一點");  
    else  
        System.out.println("猜對了");  
    return _ans == _guess;  
}  
  
}
```

Ap21.java

```
public class Ap21 {  
  
    public static void main(String[] args) {  
        GuessGame g1 = new GuessGame();  
        g1.Start();  
        do  
        {  
            g1.Guess();  
        } while (!g1.GotAns());  
    }  
}
```

**問題 22.**

輸入十個數字，排序後，由小排到大列出，並計算出合計。

解答：

Ap22.java

```
import java.util.Scanner;  
import java.util.Arrays;  
  
public class Ap01 {
```

```

public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);
    int[] ar = new int[10];

    for(int i = 0; i<10; i++)
    {
        System.out.printf("請輸入整數#%d:", i);
        ar[i] = sc.nextInt();
    }
    sc.close();
    Arrays.sort(ar);

    for(int i = 0; i<10; i++)
        System.out.print(ar[i]);
    }
}

```

**問題 23.**

高鐵票價計算，

高鐵各站票價如下

| 車站 | 南港    | 台北    | 板橋    | 桃園    | 新竹    | 苗栗    | 台中  | 彰化  | 雲林  | 嘉義    | 台南    | 左營    |
|----|-------|-------|-------|-------|-------|-------|-----|-----|-----|-------|-------|-------|
| 南港 | -     | 40    | 70    | 200   | 330   | 480   | 750 | 870 | 970 | 1,120 | 1,390 | 1,530 |
| 台北 | 40    | -     | 40    | 160   | 290   | 430   | 700 | 820 | 930 | 1,080 | 1,350 | 1,490 |
| 板橋 | 70    | 40    | -     | 130   | 260   | 400   | 670 | 790 | 900 | 1,050 | 1,320 | 1,460 |
| 桃園 | 200   | 160   | 130   | -     | 130   | 280   | 540 | 670 | 780 | 920   | 1,190 | 1,330 |
| 新竹 | 330   | 290   | 260   | 130   | -     | 140   | 410 | 540 | 640 | 790   | 1,060 | 1,200 |
| 苗栗 | 480   | 430   | 400   | 280   | 140   | -     | 270 | 390 | 500 | 640   | 920   | 1060  |
| 台中 | 750   | 700   | 670   | 540   | 410   | 270   | -   | 130 | 230 | 380   | 650   | 790   |
| 彰化 | 870   | 820   | 790   | 670   | 540   | 390   | 130 | -   | 110 | 250   | 530   | 675   |
| 雲林 | 970   | 930   | 900   | 780   | 640   | 500   | 230 | 110 | -   | 150   | 420   | 560   |
| 嘉義 | 1,120 | 1,080 | 1,050 | 920   | 790   | 640   | 380 | 250 | 150 | -     | 280   | 410   |
| 台南 | 1,390 | 1,350 | 1,320 | 1,190 | 1,060 | 920   | 650 | 530 | 420 | 280   | -     | 140   |
| 左營 | 1,530 | 1,490 | 1,460 | 1,330 | 1,200 | 1,060 | 790 | 670 | 560 | 410   | 140   | -     |

輸入起站(0-南港, 1-台北, 2-板橋....11-左營):

輸入迄站(0-南港, 1-台北, 2-板橋....11-左營):

計算票價為多少?

解答：

Ap23.java

```
import java.util.Scanner;
import java.util.Arrays;

public class Ap02 {

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int[][] price =
        {
            { 0,    40,   70,   200,  330,  480,  750,  870,  970,
              1120, 1390, 1530 },
            { 40,    0,   40,   160,  290,  430,  700,  820,  930,
              1080, 1350, 1490 },
            { 70,    40,   0,   130,  260,  400,  670,  790,  900,
              1050, 1320, 1460 },
            { 200,   160,  130,    0,  130,  280,  540,  670,  780,
              920,  1190, 1330 },
            { 330,   290,  260,   130,    0,  140,  410,  540,  640,
              790,  1060, 1200 },
            { 480,   430,  400,   280,  140,    0,  270,  390,  500,
              640,   920, 1060 },
            { 750,   700,  670,   540,  410,  270,    0,  130,  230,
              380,   650,  790 },
            { 870,   820,  790,   670,  540,  390,  130,    0,  110,
              250,   530,  675 },
            { 970,   930,  900,   780,  640,  500,  230,  110,    0,
              150,   420,  560 },
            { 1120, 1080, 1050,   920,  790,  640,  380,  250,  150,
              0,    280,  410 },
            { 1390, 1350, 1320, 1190, 1060,   920,  650,  530,  420,
              280,    0,  140 },
            { 1530, 1490, 1460, 1330, 1200, 1060,   790,  670,  560,
              410,  140,    0 }
        };

        String s;
        int n1, n2;
```

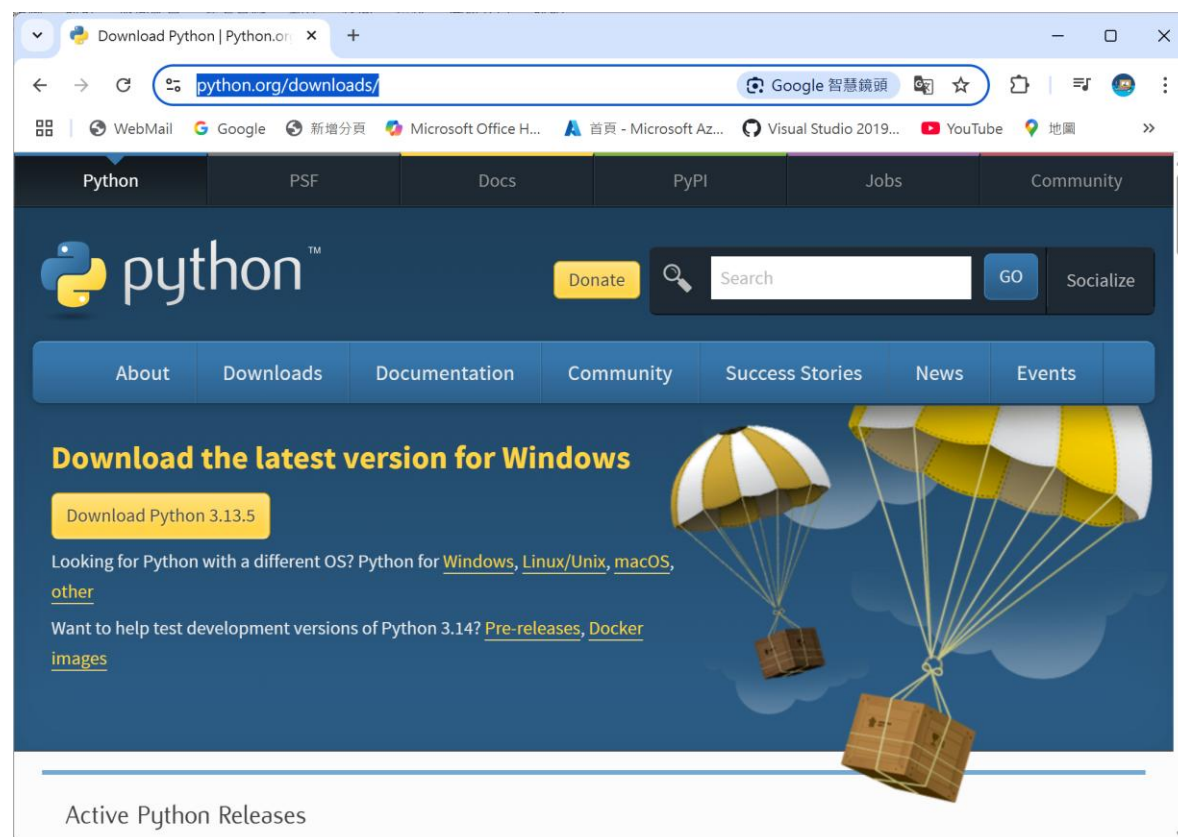


```
System.out.print("起站(0-南港/1-台北/2-板橋/3-桃園/4-新竹/  
5-苗栗/6-台中/7-彰化/8-雲林/9-嘉義/10-台南/11-左營):");  
n1 = sc.nextInt();  
  
System.out.print("迄站(0-南港/1-台北/2-板橋/3-桃園/4-新竹/  
5-苗栗/6-台中/7-彰化/8-雲林/9-嘉義/10-台南/11-左營):");  
n2 = sc.nextInt();  
sc.close();  
  
System.out.printf("票價=%d\n", price[n1][n2]);  
  
}  
}
```

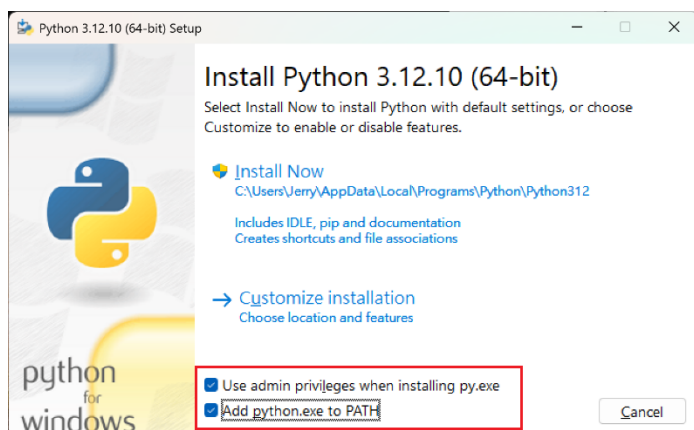
### 下載 Python

網址: <https://www.python.org/downloads/>

(Google: Python Download)



### 點選 Downloads

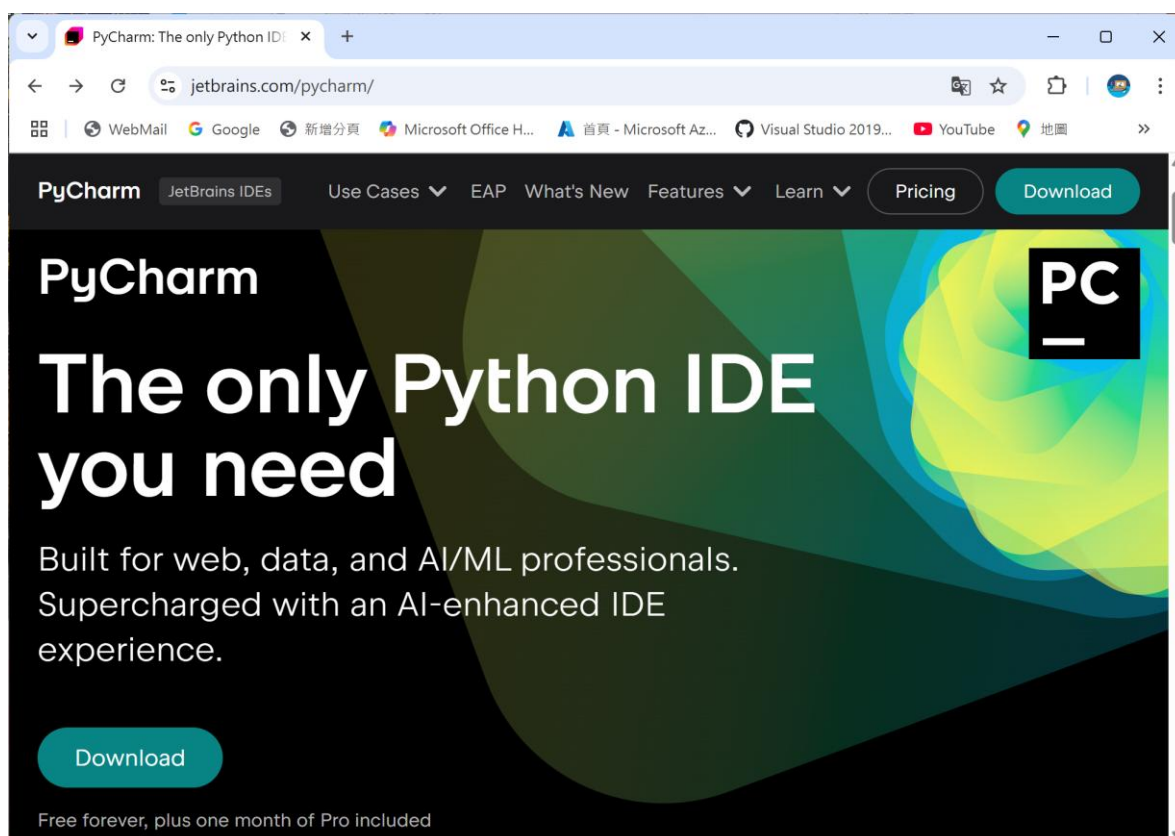


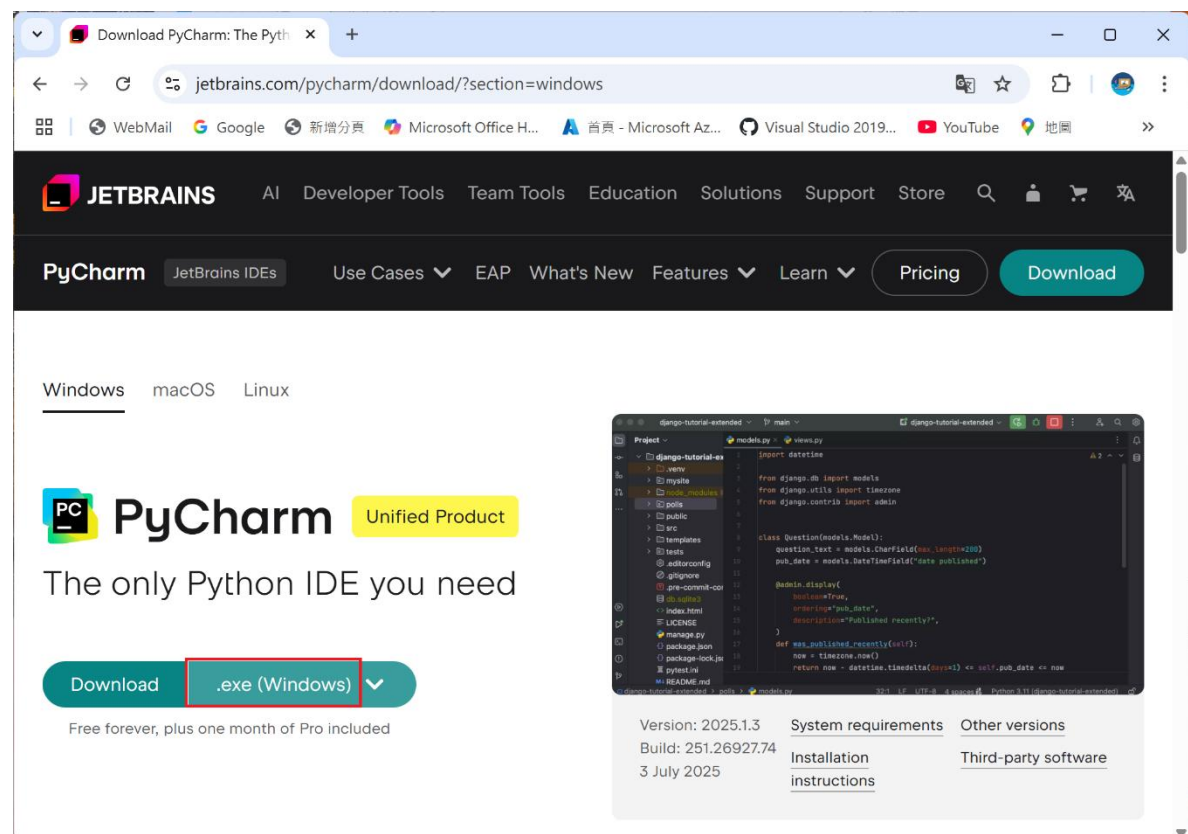
安裝 Python 時，一定要記得勾選 Add python.exe to PATH(很重要)

安裝 Python 開發工具-Pycharm

<https://www.jetbrains.com/pycharm/>

(Google: pycharm download)





### 問題 1.

Python 在畫面上顯示"Hello world!"

解答：

### 問題 2.

輸入 a, b 兩個數字，將兩數相加結果顯示在畫面上，完成偽碼(Pseudocode)以及 Python 程式碼，並執行測試結果。

解答：

#### Pseudocode

輸入 a

輸入 b

計算 a + b 的結果

#### 程式碼

```
a = int(input("請輸入 a: "))
b = int(input("請輸入 b: "))
print("a + b =", a + b)
```

**問題 3.**

輸入  $a$ ,  $b$  兩個數字，將兩數相加結果顯示在畫面上，完成偽碼(Pseudocode)以及 Python 程式碼，並執行測試結果。

解答：

**Pseudocode**

```
輸入 a
輸入 b
計算 a + b 的結果
```

**程式碼**

```
a = int(input("請輸入 a: "))
b = int(input("請輸入 b: "))
print("a + b =", a + b)
```

**問題 4.**

輸入成績，判斷是否及格(低於 60 不及格，高於或等於 60 及格)，完成偽碼(Pseudocode)以及 Python 程式碼，並執行測試結果。

解答：

**Pseudocode**

```
輸入成績(score)
IF score < 60 THEN
    顯示 “不及格”
ELSE
    顯示 “及格”
END IF
```

**程式碼**

```
score = int(input("請輸入成績: "))

if score >= 60:
    print("及格")
else:
    print("不及格")
```

問題 5.

Python 輸入十個數字，排序後，由小排到大列出，並計算出合計。

解答：

程式碼

```
numbers = [] # 用來存放使用者輸入的數字

# 依序輸入 10 個數字
for i in range(10):
    num = float(input(f"請輸入第 {i+1} 個數字："))
    numbers.append(num)

# 排序
numbers.sort()

# 計算總和
total = sum(numbers)

# 顯示結果
print("\n 排序後的數字：")
for num in numbers:
    print(num)

print(f"\n 總和：{total}")
```